

A One-Factor Model of Corporate Bond Premia

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Abstract. A one-factor model based on long-run consumption growth explains the risk premiums on corporate bond portfolios sorted on credit rating, credit spreads, downside risk, idiosyncratic volatility, long-term reversals, maturity, and sensitivity to the financial intermediary capital factor. The estimated risk-aversion coefficient is lower when we use the consumption growth of wealthy households over a longer horizon as a risk factor, and a model with a 20-quarter horizon yields a risk-aversion coefficient of 15, a value similar to the one estimated from equity portfolios.

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1. Introduction

Does macroeconomic risk explain expected returns on corporate bonds? Despite the size of the corporate bond market, there is no agreement in the literature on economic determinants of corporate bond risk premiums. In this article, we use large panel data of bond returns and show that wealthy households' consumption growth accumulated over the long horizon can explain the cross-sectional variation in risk premium.

Before examining the data, why in the first place should consumption risk explain corporate bond risk premiums? First, there is evidence in the literature that long-run consumption risk explains the aggregate credit spreads (e.g., Bhamra et al. 2010b, Chen 2010, Kuehn et al. 2023). Because credit spreads can be decomposed into risk premiums and expected losses, a model that explains credit spreads should also explain risk premiums on corporate bonds.

Second, Gilchrist and Zakrajšek (2012) empirically show that credit spreads are a strong predictor of economic growth. Thus, changes in credit spreads—roughly equivalent to bond returns—should also predict future consumption growth (which we confirm in the data), leading to comovement between bond returns and expected consumption growth. If investors have Epstein–Zin preferences, shocks to expected consumption growth carry a large price of risk, and thus, assets that comove with such shocks earn risk premiums. This evidence in the literature suggests that long-run consumption risk is a

prominent candidate that explains corporate bond risk premiums.

However, the ability of long-run consumption risk to explain bond premiums is far from obvious because of an increasingly popular view that financial intermediaries rather than households are the marginal investors who price assets. According to this view, a household's consumption growth will not align with asset returns, and thus, it does not explain risk premiums. In recent work, Haddad and Muir (2021) show that corporate credit is the most financially intermediated asset class, suggesting that consumption risk may not matter for corporate bond returns. The tension between the two priors begs extensive empirical investigation into the ability of consumption risks in explaining corporate bond risk premiums, which is the focus of our paper.

In this article, we study whether a one-factor model based on long-run consumption risk explains variation in corporate bond risk premiums associated with a wide range of bond characteristics for the period 1973–2019. Specifically, we construct seven sets of portfolios of corporate bonds sorted on credit spreads, credit rating, downside risk, idiosyncratic volatility, long-term reversal, maturity, sensitivity to the intermediary factor of He et al. (2017), and use them as test assets. To measure long-run consumption risk, we simply accumulate consumption growth up to 24 quarters and calculate the covariance between portfolio returns and consumption growth.

Our one-factor model is motivated by the long-run risk models of Bansal and Yaron (2004) and Malloy et al.

(2009). Under several simplifying assumptions, such as the unitary elasticity of intertemporal substitution (EIS), the model leads us to use cumulative consumption growth as a risk factor. We apply the generalized method of moments (GMM) estimation of the model to the data, effectively running cross-sectional regressions of average excess returns on the covariance between the factor and excess returns. The long-run risk model allows us to interpret the slope coefficient of the regression as a risk-aversion coefficient of households and to evaluate how plausible the model estimate is.

In testing the model, we use wealthy households' consumption from the Consumer Expenditure Survey (CEX) data in addition to aggregate consumption from the National Income and Product Account (NIPA) data. Although these wealthy households do not necessarily own corporate bonds directly, their consumption-savings choice is less likely to be constrained, and thus, their marginal utility is more likely to be aligned with asset returns. Therefore, with the CEX data, we can potentially measure consumption growth that is more relevant in explaining asset returns than the aggregate consumption data, which partly reflect nondiscretionary consumption.

In the data, long-run consumption risk explains a significant fraction of corporate bond risk premiums. When we use wealthy households' cumulative 20-quarter consumption growth as a risk factor, the cross-sectional R^2 is 0.80, whereas the risk-aversion coefficient is estimated at 15. This estimate is higher than the conventional level in an interval of 0.5–2 but lower than the estimates based on asset prices in the previous literature (e.g., Calvet and Czellar 2015). Notably, our estimate is similar to 10, the value used to calibrate the long-run risk model. Because the confidence interval ranges from 7.1 to 25.9, our estimate is statistically indistinguishable from 10.

Importantly, the estimated level of risk aversion is very similar to the one estimated from equities. This similarity implies that from our model's perspective, bonds and equities are priced consistently. That is, the model "solves" the credit spread puzzle in the cross-section based on the empirically estimated parameters rather than on calibrated models.

In contrast, when we use aggregate consumption growth, the estimated risk-aversion coefficient is higher; it ranges from above 250 to 50 as we expand the horizon to cumulate the growth up to 24 quarters. At the horizon with the lowest risk-aversion estimate, the cross-sectional R^2 is 0.64, lower than the estimates using wealthy households' consumption.

These results suggest that, when corporate bond returns are low, consumption tends to fall as well. Importantly, the reaction of consumption persists over the medium horizon, and it is more pronounced for wealthy households' consumption. Corporate bonds with higher average returns, such as those with higher credit spreads, predict consumption more than those with lower average

returns do. From a forward-looking investor's perspective, bonds whose returns comove with future consumption growth are regarded as risky and thus, command premiums to bear the risk. Consistent with the finding of Parker and Vissing-Jørgensen (2009), wealthy households' consumption is more sensitive to macroeconomic shocks than aggregate consumption is, and therefore, we need a lower value of risk aversion to justify the cross-sectional variation in average corporate bond returns.

In order to compare the long-run consumption risk with other factors proposed in the literature, we regress the consumption factor-mimicking portfolio excess returns on the seven factors of Fama and French (1993, 2015); the intermediary factor-mimicking portfolio of He et al. (2017); and the bond market, downside risk, and credit risk factors of Bai et al. (2019). We find that our factor earns significant alphas against the existing factors, suggesting that the long-run consumption risk is not subsumed by the existing factors.

However, we find that the consumption risk factor does not explain corporate bond portfolios sorted by the Roll illiquidity measure, bond's age, and issue size as well as betas with respect to shocks to the aggregate bond illiquidity measure proposed by Hu et al. (2013). This finding suggests that the consumption-based model is a powerful model in explaining default risk and macroeconomic uncertainty priced in corporate bonds but not necessarily risk premiums arising from illiquidity, suggesting that the consumption-based model is complementary to the intermediary-based model, which better explains risk premiums associated with liquidity.¹

Our asset pricing tests address the concerns raised by Lewellen et al. (2010) on empirical methods because we test a one-factor model using seven bond characteristics. Moreover, the performance of the long-run consumption-based model is robust to various changes in the specification of the tests. The main results hold when we (i) use the consumption of bondholders, (ii) use shocks to expectations for infinite-horizon consumption growth rate implied by a vector autoregression (VAR) as a risk factor, (iii) estimate the long-run risk model with various levels of elasticity of intertemporal substitution, (iv) allow asset returns to be not lognormally distributed, (v) use international corporate bonds as alternative test assets, and (vi) conduct asset pricing tests based on time series regressions as in Fama and French (1993) and Barillas and Shanken (2018).

In summary, we contribute to the literature by demonstrating that consumption risk explains a cross-section of corporate bond risk premiums, using the method established by Malloy et al. (2009). Our results also suggest that it is possible to tame the factor zoo for corporate bonds, as the risk premium variation associated with well-known characteristics, such as credit rating, downside risk, volatility, and long-term reversal, can be well captured by a single macroeconomic factor.

Our paper relates to the literature on the cross-section of corporate bond returns. Gebhardt et al. (2005), Jostova et al. (2013), Chordia et al. (2017), Choi and Kim (2018), Chung et al. (2019), Bali et al. (2021a, b, 2022), and Bretscher et al. (2022) document different predictors of bond returns, whereas Kelly et al. (2023) propose a reduced-form factor model. In particular, the papers closest to ours are He et al. (2017) and Bai et al. (2019, 2021). These papers present different multifactor models with different motivations to explain the cross-section of corporate bond returns. In contrast, our model uses only one risk factor and offers a more parsimonious explanation.

There is another strand of literature that incorporates consumption shocks in structural models of debt to price corporate credit spreads (e.g., Chen et al. 2008, Gourio 2013). In particular, Bhamra et al. (2010b), Chen (2010), Elkamhi and Salerno (2010), and Kuehn et al. (2023) calibrate the long-run risk models and show that the long-run risk is instrumental in explaining the aggregate credit spreads and equity risk premiums. Our study echoes the importance of the long-run risk not only for credit spreads but also for bond risk premiums.

Our paper also contributes to the literature that investigates the link between slow-moving components in consumption growth and a cross-section of asset returns, including Ait-Sahalia et al. (2004), Bansal et al. (2005, 2009), Parker and Julliard (2005), Yogo (2006), Hansen et al. (2008), Malloy et al. (2009), Bryzgalova and Julliard (2021), and Elkamhi and Jo (2023). Ferson et al. (2013) study the performance of the long-run risk model with the NIPA consumption data and conclude that it does not explain the low-grade corporate bond index well. In contrast, we use different consumption data and show that the model explains a broad cross-section of corporate bonds well.

Finally, this paper relates to the role of heterogeneous households and asset prices. Mankiw and Zeldes (1991), Basak and Cuoco (1998), Guvenen (2009), Chien et al. (2016), Lettau et al. (2019), Toda and Walsh (2019), and Elkamhi and Jo (2023) show that accounting for heterogeneity in various agents' consumption helps explain risk premiums. In this paper, we directly measure wealthy households' consumption growth instead of using indirect proxies (such as capital share) and apply it to a wide cross-section of corporate bonds.

The rest of the paper is organized as follows. In Section 2, we discuss data and the empirical application of the long-run risk model. In Section 3, we present the empirical results. In Section 4, we present several extensions of the empirical analysis, and in Section 5, we provide concluding remarks.

2. Data and Methodology

2.1. Data

2.1.1. Consumption Data. We use the consumption of wealthy households from the CEX data from March 1984 to December 2019. We calculate consumption growth in

the following way.² First, expenditures for nondurables and services from the CEX consumption categories are used to match the definition of nondurables and services in NIPA. Second, to adjust the seasonality of consumption, we regress the change in real per capita household consumption on a set of seasonal dummies and use the residual as our consumption growth measure. Finally, for each month, we compute the average consumption growth rates across wealthy households. Wealthy households are defined as the top 30% of asset holders based on their beginning of quarter holding of stocks, mutual funds, and bonds. The cutoffs for being in the top 30% are defined by the calendar year. The CEX data do not allow us to separate the ownership of bonds from other financial assets. This is not an issue for our purpose because we choose those households not because they hold corporate bonds directly but because we wish to examine the consumption of households whose intertemporal consumption decision is less likely to be constrained.

In the CEX data, a sample household is interviewed every three months over five times. Also, the CEX surveys different sets of households every month by including new households and dropping old households that finish the last interview. Therefore, we observe the quarterly consumption growth rates at a monthly frequency for the consumption of wealthy households. Section I.G in the online appendix provides more details on the CEX data and sample selection criteria.

We compute aggregate consumption growth rates using seasonally adjusted monthly real personal consumption expenditures for nondurables and services from NIPA table 2.8.3 from February 1959 to December 2019.³ Real per capita growth rates are calculated by deflating nominal values using the 2012 dollars and subtracting the log population growth rate using a monthly population from NIPA table 2.6.

Table 1 presents the summary statistics for CEX/NIPA cumulative discounted S -quarter consumption growth $\sum_{s=0}^{S-1} \delta^s (c_{t+s+1} - c_{t+s})$ with $S = 1, \dots, 20$, where the subjective discount rate δ is set to $0.95^{1/4}$. In panel A, the standard deviation of quarterly consumption growth ($S = 1$) of CEX wealthy households is 8.3%, whereas the standard deviation of the discounted cumulative 20-quarter consumption growth is 8.8%. For the NIPA aggregate consumption growth, the standard deviation is lower at 0.4% for quarterly growth but increases to 3.5% for cumulative 20-quarter growth.

For a consumption-based model to work, the factor needs to be not only volatile but also comove with asset returns. To understand the sensitivity of consumption growth to asset returns, we regress cumulative S -quarter consumption growth on quarterly returns on the value-weighted bond market portfolios. Panel B of Table 1 reports the estimated slope coefficients of the regressions. The sensitivity of wealthy households' consumption growth to bond returns is 0.26 when $S = 1$, which

Table 1. Summary Statistics of Wealthy Households' and Aggregate Consumption Growth: Volatility and Sensitivity with Respect to Asset Returns

$S =$	1	2	4	8	12	16	20	24
Panel A: Volatility of consumption growth								
CEX wealthy	0.083	0.088	0.086	0.089	0.089	0.088	0.088	0.084
CEX aggregate	0.024	0.025	0.023	0.021	0.024	0.022	0.023	0.022
NIPA aggregate	0.004	0.008	0.013	0.021	0.027	0.031	0.035	0.038
Panel B: Sensitivity to corporate bond returns								
CEX wealthy	0.260	0.370	0.253	0.098	0.145	0.450	0.383	0.258
(s.e.)	(0.130)	(0.126)	(0.173)	(0.108)	(0.132)	(0.129)	(0.114)	(0.116)
R^2	0.008	0.015	0.007	0.001	0.002	0.023	0.016	0.008
CEX aggregate	0.149	0.052	0.089	−0.012	0.027	0.044	0.037	0.037
(s.e.)	(0.044)	(0.061)	(0.054)	(0.076)	(0.057)	(0.060)	(0.074)	(0.063)
R^2	0.039	0.004	0.013	0.000	0.001	0.004	0.002	0.003
NIPA aggregate	−0.004	0.021	0.068	0.113	0.142	0.142	0.129	0.112
(s.e.)	(0.011)	(0.018)	(0.028)	(0.034)	(0.041)	(0.052)	(0.059)	(0.066)
R^2	0.001	0.012	0.042	0.042	0.041	0.031	0.021	0.014

Notes. This table reports the volatility of S -quarter growth rate of CEX wealthy households' consumption, CEX aggregate consumption, or NIPA aggregate consumption in panel A and time series regressions of those consumption measures on aggregate bond returns over different long-run horizons S in panel B,

$$\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}) = b_0 + b_1 r_{t+1} + u_{t,t+1+S},$$

where $\delta = 0.95^{1/4}$. Aggregate corporate bond returns are the value-weighted average bond returns using the bond-level data used in this study. The time period spans from April 1973 to December 2019 for NIPA data and from March 1984 to December 2019 for CEX data. Standard errors (s.e.) based on Newey and West (1987) are reported in parentheses.

increases modestly to 0.38 when $S = 20$. The sensitivity increases for a longer horizon because bond returns not only comove with consumption growth but also, predict future consumption growth. In contrast, the sensitivity of NIPA aggregate consumption growth is about 0 when $S = 1$ and increases to 0.13 when $S = 20$. Therefore, the long-run growth rate for wealthy households' consumption is three times as sensitive to aggregate shocks as the NIPA aggregate consumption growth. As explained in the next section, this high sensitivity is the key to the performance of a consumption-based model in explaining corporate bond risk premiums. Furthermore, in Online Appendix II, we study the behavior of wealthy households' consumption over the business cycle and find that wealthy households' consumption shock is more sensitive to the macroeconomic uncertainty of Jurado et al. (2015) than the aggregate consumption is.

2.1.2. Bond Return Data. For data on corporate bond returns, we mostly follow Nozawa (2017). We construct the panel data of clean prices on corporate bonds from the Lehman Brothers Fixed Income Database, The Mergent Fixed Income Securities Database (FISD)/National Association of Insurance Commissioners (NAIC) Database, Trade Reporting and Compliance Engine (TRACE), and DataStream, spanning returns from February 1973 to December 2019. For the empirical test, we use as long of time series data as possible given the availability of bond returns and consumption data, and thus, the end of the sample period is determined by the shorter of the two. For

example, in our main results where we use a 20-quarter consumption growth rate as a risk factor, our return sample ends in March 2015, which is determined by the availability of consumption data. When there are overlaps among the four databases, we prioritize them in the following order: the Lehman Brothers Fixed Income Database, TRACE, Mergent FISD/NAIC, and DataStream. Detailed descriptions of these databases as well as summary statistics of monthly returns are provided in Online Appendix I.

We remove bonds with floating rates and with option features other than callable bonds. Furthermore, we apply two filters to remove the observations that are likely to be subject to erroneous recording. First, we remove the price observations that are below \$5 or above \$1,000 per \$100 face value. Second, we remove bonds maturing in less than a year.

Based on clean prices and accrued interest in the data, we calculate a monthly return on a corporate bond in month t as

$$R_t = \frac{P_t + AI_t + \text{Coupon}_t}{P_{t-1} + AI_{t-1}} - 1, \quad (1)$$

where P_t is month t clean price, AI_t is accrued interest for the bond at the end of month t , and Coupon_t is coupon paid during month t .

Using this panel data of corporate bond returns, we form seven sets of portfolios. First, Nozawa (2017) shows that credit spreads are a strong predictor of the cross-section of corporate bond returns. Based on his finding,

He et al. (2017) test their intermediary asset pricing model using 10 portfolios sorted on credit spreads. Thus, every month, we form 10 value-weighted portfolios based on the average credit spreads between months $t - 12$ and $t - 1$. We put a one-month lag between the period where we observe the signal (the average of credit spreads) and the portfolio formation month to ensure that measurement errors in bond prices do not drive return predictability.

Next, we form portfolios sorted on credit risk, downside risk, maturity, idiosyncratic volatility, and long-term reversals. Bai et al. (2019) sort bonds based on credit rating and downside risk (measured as the 5% value at risk for bond returns over the past 36-month horizon⁴) and find that these characteristics are a strong predictor of corporate bond returns. Furthermore, Gebhardt et al. (2005) use the bond's maturity as a predictor variable, whereas Chung et al. (2019) use the idiosyncratic volatility of bond returns.⁵ More recently, Bali et al. (2021a) propose long-term reversals (measured by the negative of cumulative three-year returns from $t - 48$ to $t - 12$) as a corporate bond risk factor. Thus, every month, we sort bonds into quintiles based on these characteristics. Bai et al. (2019) also show that liquidity and short-term reversals predict corporate bond returns. Because liquidity is likely to be a different source of risk premiums than the default-related factors that we focus on, we study it separately in Section 4.4.⁶

Furthermore, we examine whether the long-run risk model can price the return spreads created by the intermediary asset pricing model of He et al. (2017). To this end, we estimate corporate bond betas against shocks to the intermediary's capital on the rolling 12-month windows.⁷ We then sort bonds every month into five value-weighted portfolios based on the preformation betas.

To summarize, we have seven sets of portfolios sorted on credit spreads, credit rating, downside risk, idiosyncratic volatility, intermediary factor betas, long-term reversal, and maturity starting around 1973 and ending in December 2019.⁸ In the empirical analysis, we use quarterly returns at the monthly frequency obtained by cumulating monthly portfolio returns.

2.2. Methodology

We follow Malloy et al. (2009) in estimating the consumption-based model. In the model, an agent in the economy has recursive preferences of a form

$$V_t = \left[(1 - \delta)C_t^{1-\frac{1}{\rho}} + \delta[E_t(V_{t+1}^{1-\frac{1}{\rho}})]^{\frac{1-\frac{1}{\rho}}{1-\gamma}} \right]^{\frac{1}{1-\gamma}}, \quad (2)$$

where C_t is consumption in month t , ρ is the elasticity of intertemporal substitution, γ is a risk-aversion coefficient, and δ is a subjective discount rate.

We assume that log consumption growth $c_{t+1} - c_t$ follows a stationary first-order VAR:

$$c_{t+1} - c_t = \mu_c + \mathbf{U}_c \mathbf{x}_t + \boldsymbol{\eta}_0 \mathbf{w}_{t+1}, \quad (3)$$

$$\mathbf{x}_{t+1} = \mathbf{G} \mathbf{x}_t + \mathbf{H} \mathbf{w}_{t+1}, \quad (4)$$

where $c_t = \log C_t$, $\mathbf{x}_t$ is a vector of a state variable that predicts consumption growth, and $\mathbf{w}_t$ is a vector of independent and identically distributed normal random variables with mean zero and covariance matrix $\mathbf{I}$. Equations (3) and (4) imply that log consumption growth can be expressed as a stationary moving-average process of the form

$$c_{t+1} - c_t = \mu_c + \sum_{s=0}^{\infty} \boldsymbol{\eta}_s \mathbf{w}_{t+1-s} \equiv \mu_c + \boldsymbol{\eta}(L) \mathbf{w}_{t+1}. \quad (5)$$

Following Hansen et al. (2008) and Malloy et al. (2009), we focus on a special case in which EIS is one in the main results. This assumption allows us to express the stochastic discount factor (SDF) derived from the preferences in (2) as a log-linear function of state variables and considerably simplifies it. We then present results when EIS differs from one as a robustness check later in Section 3.3.

Under the assumption EIS = 1, the log stochastic discount factor is

$$s_{t+1} = \log \delta - [\mu_c + \boldsymbol{\eta}(L) \mathbf{w}_{t+1}] + (1 - \gamma) \boldsymbol{\eta}(\delta) \mathbf{w}_{t+1} - \frac{1}{2} (1 - \gamma)^2 \boldsymbol{\eta}(\delta)^2, \quad (6)$$

$$= \log \delta - [c_{t+1} - c_t] + (1 - \gamma) \left[(E_{t+1} - E_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}) \right] - \frac{1}{2} (1 - \gamma)^2 \boldsymbol{\eta}(\delta)^2, \quad (7)$$

$$\approx \log \delta + (1 - \gamma) \left[(E_{t+1} - E_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}) \right] - \frac{1}{2} (1 - \gamma)^2 \boldsymbol{\eta}(\delta)^2. \quad (8)$$

In the last line, we drop the contemporaneous consumption growth $c_{t+1} - c_t$ because it is known to play a minimal role in explaining asset returns (e.g., Malloy et al. 2009). The second term in the last line is the source of a shock to the stochastic discount factor; it captures the shock to investors' expectations for the future consumption growth rate. With the Epstein–Zin preferences, such a shock affects investors' marginal utility of consumption at $t + 1$ and thus, should be reflected in asset prices.

Despite its name, the long-run risk model applies to assets with any maturity as long as their returns comove with expected consumption growth. Consider a bond with two years to maturity; this bond can command large premiums if its one-quarter return comoves well with long-run consumption growth. Thus, even though the bond matures way before the “long run,” the model can generate risk premiums for short-term assets whose returns predict consumption growth.

With a valid stochastic discount factor, the Euler equation $E[S_{t+1} R_{i,t+1}] = 1$ must hold when the agent's savings-consumption decision is unconstrained. In the main analysis, we make an additional assumption that a

return $R_{i,t+1}$ is lognormally distributed, but we relax this assumption later in Section 4. Under the lognormality assumption, we derive the unconditional Euler equation for a log return on asset i in excess of the risk-free rate of return

$$E[r_{i,t+1} - r_{f,t}] + \frac{\sigma^2(r_{i,t+1})}{2} - \frac{\sigma^2(r_{f,t})}{2} = -\text{cov}(s_{t+1}, r_{i,t+1} - r_{f,t}). \quad (9)$$

We plug the log stochastic discount factor in (8) into (9) and obtain

$$\begin{aligned} E[r_{i,t+1} - r_{f,t}] + \frac{\sigma^2(r_{i,t+1})}{2} - \frac{\sigma^2(r_{f,t})}{2} \\ \approx (\gamma - 1)\text{cov}\left((E_{t+1} - E_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}\right), \quad (10) \\ = (\gamma - 1)\text{cov}\left(\sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}\right) \\ - (\gamma - 1)\text{cov}\left(E_t \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}\right), \quad (11) \end{aligned}$$

where $r_{i,t+1}$ is a log return on an asset i and $r_{f,t}$ is the risk-free rate. In the last line, we use the law of iterated expectations to remove E_{t+1} .

Malloy et al. (2009) test a simplified version of the Euler equation in (11) by dropping the second term that captures conditional expectation for consumption growth. In this case, expected log excess returns on an asset are given by their covariance with unconditional long-run consumption growth:

$$E[r_{i,t+1} - r_{f,t}] + \frac{\sigma^2(r_{i,t+1})}{2} - \frac{\sigma^2(r_{f,t})}{2} \approx (\gamma - 1)\text{cov}\left(\sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}\right). \quad (12)$$

Equation (12) implies that, in order to obtain a low-risk aversion coefficient, we need a large covariance between asset returns and long-run consumption growth.

Malloy et al. (2009) truncate the summation over the infinite horizon in (12) up to S quarters, replacing $\sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s})$ with $\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s})$. Although this procedure involves an approximation, it has the advantage of transparency. If we test (11), we have to specify a VAR by taking a stand on the set of state variables that predict consumption growth and their dynamics. In such cases, the empirical results inevitably depend on a VAR specification and will be affected by estimation errors in a VAR model. The simplified model in (12) requires no VAR estimates because it only depends on the covariance between consumption growth and asset returns in the data. However, the simplified model has a limitation on how far we can extend the horizon S because of the

limited sample size. To strike a balance between simplicity and accuracy, we first test the simplified unconditional model in (12) with finite S and then present the conditional model in (11) that captures shocks to expected consumption growth over the infinite horizon.

When testing the Euler equation in (11) using CEX consumption data, we use the average of log consumption growth across wealthy households, such that $c_{t+1} - c_t = (1/H_t) \sum_h (c_{h,t+1} - c_{h,t})$. This procedure implies that we start from the Euler equation for each household and then aggregate it across households without using the representative agent framework.

Wealthy households typically do not own corporate bonds directly but hold them through mutual funds (Bai et al. 2021). This does not automatically invalidate the link between their consumption growth and bond risk premiums. In classic portfolio theories, optimal holdings can be described as a combination of several portfolios (such as a tangency portfolio and a risk-free asset) under certain conditions,⁹ which is called the “mutual fund” separation theorems. If these funds are mean-variance efficient, then their returns and owners’ consumption can serve as a valid risk factor irrespective of whether the households own bonds directly or through mutual funds. Importantly, when we examine the consumption risk of bonds, we do not assume that wealthy households’ consumption decision “moves” corporate bond prices through fire sales and purchases. Rather, we view unobservable macroeconomic shocks as affecting both households’ consumption decisions and bond returns simultaneously, and we use the resulting comovement as a measure of risk.

3. Empirical Results

3.1. Measuring Long-Run Consumption Risk

Before estimating the Euler equation in (12), we dissect the long-run consumption risk of corporate bond portfolios using the covariance between long-run consumption growth and quarterly returns on bond portfolios. To understand why the model works, we compare long-run risk with short-run risk:

$$\begin{aligned} \underbrace{\text{cov}\left(\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1}^e\right)}_{\text{Long run}} &= \underbrace{\text{cov}(c_{t+1} - c_t, r_{i,t+1}^e)}_{\text{Short run}} \\ &+ \text{cov}\left(\sum_{s=1}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1}^e\right), \quad (13) \end{aligned}$$

where $r_{i,t+1}^e = r_{i,t+1} - r_{f,t}$.

The short-run risk corresponds to contemporaneous covariance between consumption growth and returns, which would be the only source of priced risk for a model with power utility households (e.g., Lucas 1978). The difference between the long-run and short-run risks

comes from the predictability of future consumption growth with bond returns. If a higher return in quarter $t + 1$ predicts higher consumption growth afterward, then this long-run risk is greater than the short-run risk.¹⁰

Tables 2 and 3 present average excess returns, short-run covariance $\hat{cov}(c_{t+1} - c_t, r_{i,t+1} - r_{f,t})$, and long-run covariance measured by the discounted 20-quarter cumulative consumption growth of CEX wealthy households $\hat{cov}(\sum_{s=0}^{19} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}^f)$ for the 40 portfolios we use as test assets. For all sorting variables, the long-run risk model generates a similar pattern in covariance as that in average excess returns. Specifically, the covariance with 20-quarter consumption growth increases nearly monotonically from the first quintile/decile to the last quintile/decile for all subsets of portfolios. The average excess returns for those portfolios increase similarly, suggesting that the long-run risk model not only generates large risk exposure but also, the pattern in risk exposure that matches that in average excess returns. Moreover, the difference in long-run covariance between the last and first quintile/decile is larger than that in short-run covariance, underscoring the importance of long-run consumption risk.

Figure 1(a) visualizes the information in Tables 2 and 3, plotting average excess returns on the y axis and covariance on the x axis. The covariance is greater when we use wealthy households' consumption growth over the long horizon than short-run consumption growth, and thus, the fitted line flattens. This implies that the return spread between the last and first quintile/decile predicts their consumption growth. Figure 1(b) shows the same plot but using NIPA aggregate consumption growth over one and eight quarters instead of CEX wealthy households' consumption. We treat eight quarters as a long horizon

here because as we show, extending the horizon further does not flatten the fitted line for the aggregate consumption growth. Even though using a long-run growth rate helps lower the slope, the fitted line is still much steeper than that in panel (a), which suggests that we need a higher risk-aversion coefficient to rationalize the observed variation in average returns if we use the aggregate consumption growth rate.

In panel (a), we use a shorter sample from 1984 to 2019 because of the availability of the CEX data, starting when the treasury yield hit the postwar record high and ending when the yield is near zero. As a result, for most portfolios, the average returns in excess of T-bill rates are higher in panel (a) than in panel (b), which starts from 1973. The implication of the unique sample period for the CEX data will be discussed in the next section.

3.2. Fit of the Long-Run Risk Model

3.2.1. Estimation Using All Test Assets. In this section, we estimate the parameters of the long-run risk model in (12) and formally evaluate the performance of the model in explaining the cross-section of corporate bond premiums. Following Malloy et al. (2009), we use the GMM framework to simultaneously estimate covariance between long-run consumption growth and asset returns together with model parameters.¹¹ For the test assets, we use 40 corporate bond portfolios sorted on seven characteristics described in Tables 2 and 3. Given the critique of Lewellen et al. (2010), it is important to include multiple test assets in the test to break a tight factor structure in returns on the test assets; if we include only one set of portfolios based on a univariate sort, then the resulting cross-sectional fit can be mechanical.

Table 2. Average Excess Returns and Short-Run/Long-Run Covariances Between Excess Returns and Wealthy Households' Consumption Growth

	Low									High	High – low
	1	2	3	4	5	6	7	8	9	10	10 – 1
Credit spread portfolios											
Returns (%)	0.8219 (0.1104)	0.9969 (0.1191)	1.0989 (0.1529)	1.0845 (0.1680)	1.1634 (0.1834)	1.2405 (0.1688)	1.2518 (0.1916)	1.3210 (0.1883)	1.3151 (0.2475)	2.4959 (0.4835)	1.6740 (0.4688)
SR (%)	0.0091 (0.0047)	0.0217 (0.0085)	0.0199 (0.0105)	0.0261 (0.0092)	0.0327 (0.0091)	0.0322 (0.0090)	0.0263 (0.0100)	0.0278 (0.0088)	0.0149 (0.0113)	0.0504 (0.0173)	0.0413 (0.0188)
LR (%)	0.0186 (0.0063)	0.0219 (0.0075)	0.0197 (0.0091)	0.0226 (0.0082)	0.0258 (0.0088)	0.0226 (0.0077)	0.0232 (0.0074)	0.0296 (0.0076)	0.0399 (0.0146)	0.1115 (0.0368)	0.0929 (0.0379)

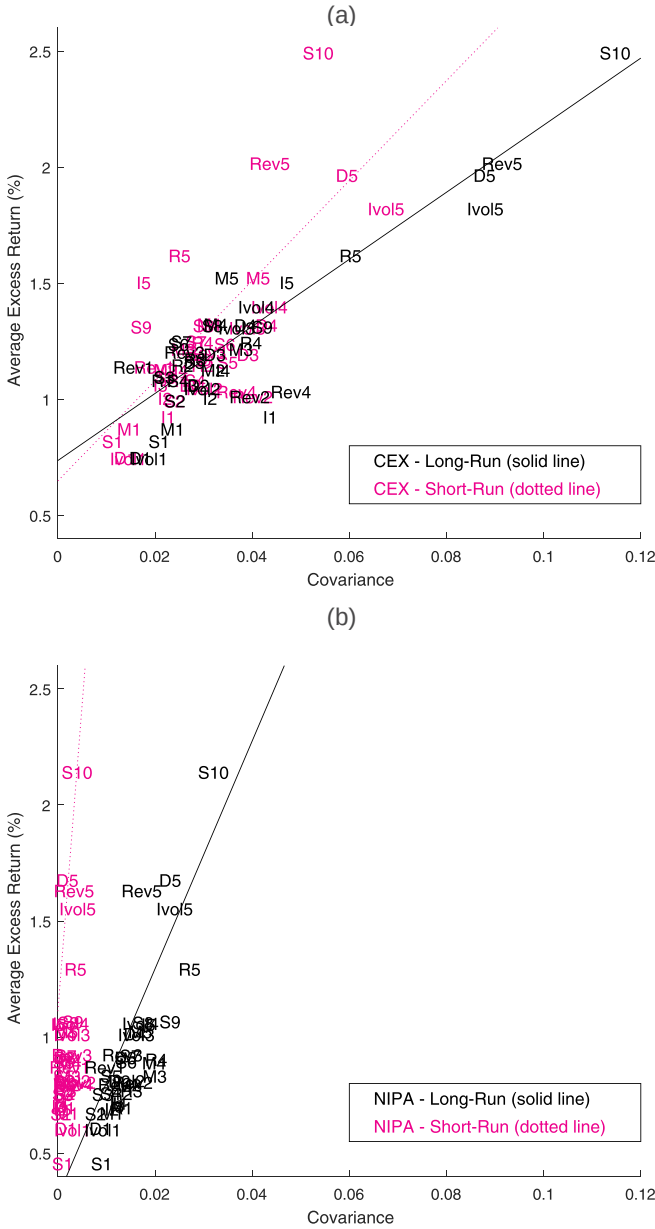
Notes. This table reports average excess returns, $\hat{E}(r_{i,t+1} - r_{f,t}) + (\hat{\sigma}^2(r_{i,t+1})/2 - (\hat{\sigma}^2(r_{f,t}))/2)$, and the quantity of risk for the short-run risk (denoted by SR), $\hat{cov}(c_{t+1} - c_t, r_{i,t+1} - r_{f,t})$, and the long-run risk (denoted by LR), $\hat{cov}(\sum_{s=0}^{19} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}^f)$, for each portfolio group where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative 20-quarter consumption growth. Consumption of wealthy households defined as the top 30% of asset holders from CEX data is used. The covariances are computed as the unconditional sample covariances. Bootstrapped standard errors computed with 5,000 replications are reported in parentheses. The time period spans from March 1984 to December 2019.

Table 3. Average Excess Returns and Short-Run/Long-Run Covariances Between Excess Returns and Wealthy Households' Consumption Growth

	Low				High	High – low
	1	2	3	4	5	5 – 1
Panel A: Downside portfolios						
Returns (%)	0.7512 (0.0531)	1.0641 (0.1282)	1.1965 (0.1872)	1.3227 (0.2378)	1.9685 (0.3919)	1.2173 (0.3580)
SR (%)	0.0115 (0.0043)	0.0250 (0.0075)	0.0368 (0.0104)	0.0407 (0.0118)	0.0572 (0.0147)	0.0457 (0.0125)
LR (%)	0.0147 (0.0046)	0.0266 (0.0059)	0.0300 (0.0084)	0.0363 (0.0092)	0.0855 (0.0232)	0.0709 (0.0231)
Panel B: Maturity portfolios						
Returns (%)	0.8750 (0.0683)	1.1297 (0.1334)	1.2188 (0.2083)	1.3261 (0.1960)	1.5266 (0.2410)	0.6516 (0.2008)
SR (%)	0.0122 (0.0037)	0.0196 (0.0062)	0.0233 (0.0090)	0.0286 (0.0126)	0.0387 (0.0145)	0.0265 (0.0118)
LR (%)	0.0211 (0.0048)	0.0294 (0.0071)	0.0352 (0.0105)	0.0300 (0.0084)	0.0323 (0.0116)	0.0112 (0.0076)
Panel C: Rating portfolios						
Returns (%)	1.0844 (0.1177)	1.1498 (0.1959)	1.1757 (0.1619)	1.2472 (0.1852)	1.6225 (0.3221)	0.5381 (0.2677)
SR (%)	0.0223 (0.0098)	0.0275 (0.0099)	0.0270 (0.0102)	0.0275 (0.0088)	0.0227 (0.0141)	0.0004 (0.0102)
LR (%)	0.0192 (0.0077)	0.0233 (0.0079)	0.0259 (0.0104)	0.0374 (0.0082)	0.0579 (0.0211)	0.0387 (0.0199)
Panel D: Intermediary portfolios						
Returns (%)	0.9271 (0.2751)	1.0075 (0.1038)	1.0647 (0.1103)	1.1280 (0.1354)	1.5068 (0.3014)	0.5797 (0.2505)
SR (%)	0.0212 (0.0081)	0.0205 (0.0069)	0.0196 (0.0079)	0.0238 (0.0077)	0.0162 (0.0149)	–0.0050 (0.0089)
LR (%)	0.0422 (0.0112)	0.0298 (0.0078)	0.0257 (0.0069)	0.0325 (0.0066)	0.0456 (0.0176)	0.0034 (0.0092)
Panel E: Idiosyncratic portfolios						
Returns (%)	0.7479 (0.0530)	1.0495 (0.1392)	1.3109 (0.1861)	1.4016 (0.2480)	1.8245 (0.3492)	1.0767 (0.3215)
SR (%)	0.0107 (0.0041)	0.0264 (0.0075)	0.0352 (0.0104)	0.0397 (0.0119)	0.0638 (0.0158)	0.0531 (0.0147)
LR (%)	0.0150 (0.0044)	0.0259 (0.0060)	0.0330 (0.0075)	0.0371 (0.0097)	0.0842 (0.0209)	0.0692 (0.0215)
Panel F: Long-term reversal portfolios						
Returns (%)	1.1421 (0.2137)	1.0146 (0.1526)	1.2065 (0.1174)	1.0361 (0.2170)	2.0200 (0.3402)	0.8779 (0.3332)
SR (%)	0.0156 (0.0118)	0.0359 (0.0088)	0.0262 (0.0093)	0.0326 (0.0086)	0.0394 (0.0095)	0.0238 (0.0102)
LR (%)	0.0114 (0.0103)	0.0353 (0.0085)	0.0219 (0.0096)	0.0438 (0.0085)	0.0872 (0.0188)	0.0759 (0.0195)

Notes. This table reports average excess returns, $\hat{E}(r_{i,t+1} - r_{f,t}) + (\hat{\sigma}^2(r_{i,t+1}))/2 - (\hat{\sigma}^2(r_{f,t}))/2$, and the quantity of risk for the short-run risk (denoted by SR), $\hat{cov}(c_{t+1} - c_t, r_{i,t+1} - r_{f,t})$, and the long-run risk (denoted by LR), $\hat{cov}(\sum_{s=0}^{19} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}^f)$, for each portfolio group where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative 20-quarter consumption growth. Consumption of wealthy households defined as the top 30% of asset holders from CEX data is used. The covariances are computed as the unconditional sample covariances. Bootstrapped standard errors computed with 5,000 replications are reported in parentheses. The time period spans from March 1984 to December 2019.

Figure 1. (Color online) Average Bond Returns and Covariances Between Bond Returns and Consumption Growth



Notes. (a) CEX wealthy household consumption (March 1984 to December 2019). (b) NIPA aggregate consumption (February 1973 to December 2019). This figure plots the average bond excess returns (quarterly) against covariances of excess returns with the short-run ($c_{t+1} - c_t$) or long-run consumption growth rate ($\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s})$), where $S = 20$ quarters for CEX and $S = 8$ quarters for NIPA). Dotted lines are from an Ordinary Least Squares (OLS) regression.

Effectively, we run cross-sectional regressions

$$\begin{aligned} \hat{E}[r_{i,t+1} - r_{f,t}] + \frac{\hat{\sigma}^2(r_{i,t+1})}{2} - \frac{\hat{\sigma}^2(r_{f,t})}{2} \\ = \zeta + (\gamma - 1)\hat{\sigma}_{i,c} + e_i, \end{aligned} \quad (14)$$

$$\hat{\sigma}_{i,c} = \hat{cov}\left(\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}\right), \quad (15)$$

where $r_{i,t+1}$ is a log return on an asset i and $r_{f,t}$ is the log 30-day T-bill rate. To evaluate the fit of the model, we define the cross-sectional $\bar{R}^2$ by $\bar{R}^2 = 1 - (\text{Var}_c(\hat{E}[R_i^e] - \hat{R}_i^e) / \text{Var}_c(\hat{E}[R_i^e]))$, where $\hat{E}[R_i^e] = \hat{E}[r_{i,t+1} - r_{f,t}] + (\hat{\sigma}^2(r_{i,t+1})/2 - (\hat{\sigma}^2(r_{f,t}))/2)$ and $\hat{R}_i^e$ is the fitted value. To calculate the standard errors and 95% confidence intervals for estimates and $\bar{R}^2$, we conduct bootstrap simulations by randomly drawing months with replacement 5,000 times, which accounts for the cross-sectional correlation in asset returns.¹²

Table 4 presents the main results of the article, including the estimated coefficients in (14) and cross-sectional $\bar{R}^2$ statistics with different values of S . In panel A, the factors are based on CEX wealthy households' consumption, whereas in panel B, they are from NIPA aggregate consumption. To evaluate the importance of long-run risk, we show the estimates in (14) with different values of S in the third to last column in panel A of Table 4. Starting from the results using the consumption of wealthy households in CEX, we find that the risk-aversion estimates are lower at the medium horizon than for the short horizon. For example, when $S = 1$ quarter, γ is estimated at 24, and when $S = 20$ quarters, the estimate yields a reasonable level of risk aversion of $\gamma = 15$. These results suggest that bond returns predict the wealthy households' consumption growth better over the medium term, which leads to a greater quantity of risk. As a result, the estimated risk aversion becomes lower as we increase S up to 20 quarters. When $S = 20$, the 95% bootstrapped confidence interval ranges from 7.1 to 25.9. Therefore, the risk-aversion coefficient is not statistically distinguishable from 10—a commonly used value in the long-run risk model (e.g., Bansal and Yaron 2004)—but they are significantly different from zero, supporting the validity of the estimate.¹³

With regard to the explanatory power of the model, when $S = 1$ quarter, the cross-sectional $\bar{R}^2$ using wealthy households' consumption growth is 0.33. However, when $S = 20$ quarters, the cross-sectional $\bar{R}^2$ increases to 0.80, which is substantial given that we use a one-factor model on seven sets of portfolios and higher than the value reported in Malloy et al. (2009).¹⁴ The cross-sectional $\bar{R}^2$ does not monotonically increase as S increases, owing to the relatively large measurement errors in the survey-based consumption growth rate. In Section 3.3, we address potential concerns arising from the noise in the data by confirming that similar results hold with $S = \infty$ using a VAR.

In contrast, if we estimate a model with NIPA aggregate consumption with $S = 1$ quarter, the estimated γ is 255, which is implausibly high but consistent with those found in the literature using contemporaneous aggregate consumption growth (e.g., 365 in Nagel and Singleton 2011). Similar to the results using the CEX data, as S increases, the estimated risk-aversion parameter decreases

Table 4. GMM Cross-Sectional Regression Using All Corporate Bond Portfolios

S (quarters)	1	2	4	8	12	16	20	24
Regressions: $\hat{E}[r_{i,t+1} - r_{f,t}] + \frac{\hat{\sigma}^2(r_{i,t+1})}{2} - \frac{\hat{\sigma}^2(r_{f,t})}{2} = \zeta + (\gamma - 1)c\hat{v} \left(\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t} \right) + e_i$								
Panel A: CEX consumption of wealthy households								
ζ (%)	0.71 [0.46 1.29]	0.46 [0.12 1.20]	0.85 [0.43 1.23]	0.99 [0.60 1.41]	0.96 [0.50 1.69]	0.55 [0.24 0.93]	0.74 [0.42 0.96]	0.73 [0.24 1.10]
γ	23.5 [−2.3 40.7]	23.6 [0.5 35.1]	17.1 [−4.7 31.6]	21.9 [−20.0 45.7]	16.8 [−20.5 39.1]	16.1 [4.3 25.9]	15.4 [7.1 25.9]	23.5 [5.8 46.5]
$\bar{R}^2$	0.33 [0.00 0.67]	0.72 [0.00 0.93]	0.21 [0.00 0.75]	0.29 [0.00 0.66]	0.13 [0.00 0.55]	0.69 [0.05 0.90]	0.80 [0.26 0.90]	0.62 [0.08 0.80]
$\frac{RMSE}{RMSR}$	0.22	0.14	0.25	0.24	0.26	0.14	0.12	0.17
Number of assets	40	40	40	40	40	40	40	40
Number of asset-month	16,940	16,820	16,580	16,100	15,620	15,140	14,660	14,180
Panel B: NIPA aggregate consumption								
ζ (%)	0.63 [0.23 1.00]	0.33 [−0.06 1.01]	0.29 [−0.06 1.02]	0.26 [−0.17 1.01]	0.18 [−0.17 0.97]	0.24 [−0.14 1.10]	0.30 [−0.07 1.18]	0.78 [0.19 1.29]
γ	254.6 [119.4 332.5]	130.7 [35.9 188.6]	71.0 [17.5 101.8]	50.1 [17.1 67.9]	50.4 [12.8 69.9]	56.8 [−24.3 74.8]	78.1 [−56.0 92.3]	29.6 [−71.0 83.5]
$\bar{R}^2$	0.67 [0.30 0.78]	0.57 [0.08 0.74]	0.59 [0.05 0.76]	0.64 [0.09 0.79]	0.62 [0.08 0.82]	0.58 [0.05 0.81]	0.53 [0.00 0.82]	0.04 [0.00 0.76]
$\frac{RMSE}{RMSR}$	0.18	0.21	0.21	0.20	0.20	0.19	0.22	0.32
Number of assets	40	40	40	40	40	40	40	40
Number of asset-month	21,400	21,280	21,040	20,560	20,080	19,600	19,120	18,640
Panel C: CEX consumption of wealthy households using matching treasuries as risk-free assets								
ζ (%)	0.31 [−0.12 0.53]	−0.11 [−0.28 0.25]	0.08 [−0.27 0.29]	0.02 [−0.22 0.33]	0.17 [−0.15 0.39]	−0.12 [−0.32 0.16]	−0.11 [−0.28 0.12]	−0.01 [−0.23 0.31]
γ	17.1 [−24.7 45.8]	19.1 [−17.6 32.1]	11.4 [−11.0 33.1]	23.3 [−21.1 49.8]	12.2 [−29.1 37.6]	12.4 [0.1 23.3]	14.0 [5.6 21.4]	21.9 [0.7 47.4]
$\bar{R}^2$	0.09 [0.00 0.55]	0.43 [0.00 0.75]	0.09 [0.00 0.65]	0.35 [0.00 0.76]	0.07 [0.00 0.55]	0.41 [0.00 0.75]	0.73 [0.11 0.85]	0.54 [0.01 0.81]
$\frac{RMSE}{RMSR}$	0.77	0.62	0.79	0.66	0.82	0.70	0.46	0.58
Number of assets	40	40	40	40	40	40	40	40
Number of asset-month	16,940	16,820	16,580	16,100	15,620	15,140	14,660	14,180
Panel D: CEX aggregate consumption								
ζ (%)	0.44 [0.26 1.22]	0.79 [0.51 1.32]	0.44 [0.21 1.22]	1.02 [0.64 1.41]	0.52 [0.27 1.27]	0.43 [0.23 1.39]	0.83 [0.33 1.36]	0.92 [0.11 1.49]
γ	60.7 [−23.6 82.1]	85.6 [−53.6 102.4]	75.8 [−25.7 103.3]	71.2 [−43.3 95.3]	104.7 [−41.4 118.9]	108.4 [−56.4 118.8]	64.4 [−40.7 90.3]	51.3 [−65.6 119.7]
$\bar{R}^2$	0.61 [0.00 0.89]	0.66 [0.01 0.83]	0.48 [0.00 0.89]	0.70 [0.00 0.85]	0.79 [0.01 0.92]	0.50 [0.00 0.88]	0.31 [0.00 0.88]	0.08 [0.00 0.91]
$\frac{RMSE}{RMSR}$	0.17	0.16	0.20	0.15	0.13	0.18	0.22	0.27
Number of assets	40	40	40	40	40	40	40	40
Number of asset-month	16,940	16,820	16,580	16,100	15,620	15,140	14,660	14,180
Panel E: CEX consumption of bondholders								
ζ (%)	0.85 [0.68 1.36]	0.64 [0.47 1.14]	0.52 [0.31 1.13]	0.98 [0.73 1.34]	0.80 [0.44 1.47]	0.50 [0.3 1.03]	0.82 [0.44 1.16]	0.68 [0.41 1.29]
γ	26.3 [1.6 38.9]	20.7 [3.3 30.2]	21.6 [−0.5 27.7]	23.9 [−19.3 34.2]	28.9 [−15.4 41.8]	19.6 [1.2 28.7]	14.9 [6.6 24.1]	22.0 [1.7 28.7]
$\bar{R}^2$	0.44 [0.01 0.66]	0.86 [0.01 0.93]	0.69 [0.00 0.90]	0.67 [0.00 0.83]	0.56 [0.00 0.84]	0.79 [0.01 0.90]	0.80 [0.15 0.91]	0.68 [0.02 0.85]

Table 4. (Continued)

S (quarters)	1	2	4	8	12	16	20	24
$\frac{RMSE}{RMSR}$	0.20	0.10	0.15	0.16	0.18	0.11	0.12	0.16
Number of assets	40	40	40	40	40	40	40	40
Number of asset-month	16,940	16,820	16,580	16,100	15,620	15,140	14,660	14,180

Notes. This table reports GMM cross-sectional regression results over different long-run horizons S : $\hat{E}[r_{i,t+1} - r_{f,t}] + (\hat{\sigma}^2(r_{i,t+1}) - \hat{\sigma}^2(r_{f,t}))/2 = \zeta + (\gamma - 1)\hat{c}\hat{v}(\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}) + e_i$, where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill in panels A, B, D, and E, whereas it is the log return on matching treasury bonds in panel C, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative consumption growth over multiple horizons $\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s})$. Panels A, C, and E report the results using wealthy households (defined as the top 30% of asset holders) from CEX, whereas panels B and D report those using the consumption growth of aggregate households from NIPA and from CEX, respectively. Panel E reports the results using the consumption of CEX corporate bondholders. The quantity of risk is jointly estimated with parameters ζ , η , and γ using GMM. Test assets are 40 portfolios including 10 credit spread-sorted portfolios, 5 downside risk-sorted portfolios, 5 maturity-sorted portfolios, 5 credit rating-sorted portfolios, 5 intermediary factor (He et al. 2017) beta-sorted portfolios, 5 idiosyncratic volatility-sorted portfolios, and 5 long-term reversal portfolios. Reported are the intercepts ζ , η , and implied risk-aversion coefficient γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \bar{R}_i^e) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\bar{R}_i^e$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N)\sum_{i=1}^N (E(R_i^e) - \bar{R}_i^e)^2}$ and $RMSR = \sqrt{(1/N)\sum_{i=1}^N E(R_i^e)^2}$. The time period spans from March 1984 to December 2019 for CEX and from February 1973 to December 2019 for NIPA. Unconditional pricing errors ζ and η are multiplied by 100 for ease of exposition.

up to a certain level of S . When $S = 2$ quarters, the estimated γ goes down to 131 and reaches the bottom when $S = 8$ and the estimated γ is 50. The cross-sectional $\bar{R}^2$ statistics remain similar, as it is 0.67 when $S = 1$ and 0.64 when $S = 8$, suggesting that the model using the NIPA aggregate consumption fits the data reasonably well but not as good as wealthy households' consumption does.

The better performance for the wealthy households' consumption risk than the aggregate consumption partly reflects a variety of measurement issues for the aggregate data raised in the previous work (e.g., Ait-Sahalia et al. 2004, Savov 2011, Kroencke 2017). Furthermore, aggregate consumption may not capture well discretionary consumption-savings decisions that are tied to asset returns.¹⁵ As a result, the consumption of wealthy households lines up better with observed asset returns than the aggregate consumption does.¹⁶

The intercept of the cross-sectional regression for wealthy households' 20-quarter consumption growth is 0.74% per quarter, suggesting that on average corporate bonds have returns that are too high during the sample period, whereas the intercept for the NIPA consumption growth is insignificant at 0.30%. This mispricing of the risk-free asset can be explained by the strong downward trend in treasury yields over the sample period for the CEX data.

To account for the impact of falling treasury yields, we repeat the exercise using corporate bond returns in excess of matched treasury bonds with the same cash flows. To this end, we create synthetic treasury securities with the same coupon and maturity as each corporate bond in the sample¹⁷ and use their returns as the risk-free rate of returns in (14). Because corporate and matching treasury bonds have the same interest rate risk, this

excess return is in principle not affected by shifts in the treasury yield curve. Panel C of Table 4 reports the estimates for the Euler equation using CEX wealthy household consumption as a risk factor. We find that the estimated risk-aversion coefficient and the cross-sectional $\bar{R}^2$ remain nearly unchanged from our main results in panel A. For example, for $S = 20$, risk aversion is estimated at 14 with $\bar{R}^2$ of 0.73. However, the intercept falls to -0.11% , which is statistically indistinguishable from zero. Therefore, once we account for changing treasury yields, the wealthy households' consumption risk can price both corporate bond portfolios and risk-free assets simultaneously.

The better performance using wealthy households' consumption is not driven by the difference in data quality between NIPA and CEX. In panel D of Table 4, we repeat the GMM estimation using CEX consumption using all households (rather than the top 30% of asset holders). We find that when $S = 20$, the estimated risk-aversion coefficient is 64, higher than the estimate based on the NIPA aggregate consumption (50 with $S = 8$), whereas the cross-sectional $\bar{R}^2$ is 0.31. Thus, the success of the consumption-based model is not because of the usage of the CEX data per se. Rather, it is driven by the fact that we focus on wealthy households' consumption instead of aggregate consumption. In Tables IA3 and IA4 in the online appendix, we show that our results are not sensitive to the particular choice of the cutoff (30%) to define wealthy households.

The evidence suggests that wealthy households' consumption prices the cross-section of corporate bonds well. These results are consistent with the model's performance on equity, as shown in Malloy et al. (2009). To

verify, in Table 5, we estimate the model using the Fama–French 25 portfolios as test assets over the sample period up to 2019, extending the sample in the original study by 15 years.¹⁸ With the quarterly consumption growth, the risk-aversion coefficients estimated from equities are 18 (CEX wealthy) and 166 (NIPA aggregate), lower than the corresponding values estimated from bonds (24 and 255). However, when we use 20-quarter wealthy households' consumption growth, the risk-aversion coefficient inferred from equity is 17, which is remarkably similar to 15, the value estimated from corporate bonds. Thus, the corporate bond risk premiums appear too high relative to the equity counterpart if we use the short-term consumption growth as a risk factor. This tension, known as the credit spread puzzle, is often attributed to features specific to bond cash flows, such as idiosyncratic tail risks (Culp et al. 2018). We resolve the tension by showing that such risks are associated with the long-term consumption growth of wealthy households; once a left-tail event realizes, it disproportionately

affects wealthy households' consumption in the future. Once we account for this link, we can build a one-factor model with a risk-aversion parameter that is consistent with both equity and bond risk premiums.

Although our focus is on wealthy households' consumption, we also attempt to estimate bondholders' consumption motivated by the idea that the SDF is a function of the consumption of market participants. Because of the lack of data in CEX on whether households own bonds or not, we use a sample of the Survey of Consumer Finances, which indicates bondholders, and identify key characteristics that predict corporate bond ownership by estimating a probit model. The details on the probit analysis that links households' characteristics and bond ownership are provided in Online Appendix IV. We then use the estimated coefficients to calculate the probability of bond ownership in the CEX sample and create the consumption growth of households that have at least a 10% chance of holding corporate bonds. We call this series bondholders' consumption for short.

Table 5. GMM Cross-Sectional Regression Using Equities

S (quarters) =	1	2	4	8	12	16	20	24
Panel A: CEX consumption of wealthy households								
ζ (%)	2.59 [1.46 3.56]	1.60 [0.60 3.29]	2.23 [1.32 3.28]	2.30 [1.45 3.40]	2.72 [1.79 3.50]	2.14 [1.34 3.58]	1.69 [0.84 3.15]	2.50 [1.71 3.64]
γ	18.0 [−15.3 26.8]	16.8 [−6.4 24.5]	13.3 [0.7 25.7]	20.5 [1.2 25.5]	11.2 [−6.7 23.6]	10.7 [−8.0 19.2]	17.3 [2.1 24.5]	20.6 [−14.6 25.8]
$\bar{R}^2$	0.27 [0.00 0.48]	0.35 [0.00 0.49]	0.22 [0.00 0.46]	0.55 [0.01 0.68]	0.10 [0.00 0.56]	0.10 [0.00 0.45]	0.43 [0.00 0.58]	0.31 [0.00 0.60]
$\frac{RMSE}{RMSR}$	0.18	0.17	0.19	0.15	0.20	0.20	0.16	0.18
Number of assets	25	25	25	25	25	25	25	25
Number of asset-month	10,750	10,675	10,525	10,225	9,925	9,625	9,325	9,025
Panel B: NIPA aggregate consumption								
ζ (%)	0.05 [−0.99 3.80]	0.39 [−1.07 3.99]	1.91 [0.02 3.63]	0.93 [0.01 3.06]	1.48 [0.11 2.67]	1.86 [0.69 2.78]	1.68 [0.70 2.79]	1.75 [0.99 2.91]
γ	165.7 [−129.1 242.4]	105.4 [−82.7 171.0]	30.4 [−38.6 98.6]	46.4 [−9.8 80.5]	37.5 [7.4 81.2]	31.5 [6.8 85.8]	29.3 [3.7 74.4]	28.3 [0.6 61.8]
$\bar{R}^2$	0.50 [0.05 0.69]	0.31 [0.00 0.62]	0.07 [0.00 0.55]	0.40 [0.00 0.72]	0.43 [0.04 0.72]	0.34 [0.03 0.69]	0.34 [0.01 0.67]	0.31 [0.01 0.61]
$\frac{RMSE}{RMSR}$	0.15	0.18	0.21	0.17	0.16	0.17	0.18	0.18
Number of assets	25	25	25	25	25	25	25	25
Number of asset-month	18,225	18,150	18,000	17,700	17,400	17,100	16,800	16,500

Notes. This table reports GMM cross-sectional regression results with different long-run horizons S using equity portfolios as test assets: $\hat{E}[r_{i,t+1} - r_{f,t}] + (\sigma^2(r_{i,t+1}))/2 - (\sigma^2(r_{f,t}))/2 = \zeta + (\gamma - 1)c\bar{o}v(\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}) + e_i$, where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative consumption growth over multiple horizons $\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s})$. The quantity of risk is jointly estimated with parameters ζ and γ using GMM. Test assets are 25 Fama–French size- and book-to-market-sorted portfolios from July 1926 to December 2019. Reported are the intercepts ζ and implied risk-aversion coefficients γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \hat{R}_i^e) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\hat{R}_i^e$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N) \sum_{i=1}^N (E(R_i^e) - \hat{R}_i^e)^2}$ and $RMSR = \sqrt{(1/N) \sum_{i=1}^N E(R_i^e)^2}$. The time period spans from March 1984 to December 2019 for CEX wealthy households' consumption and from April 1959 to December 2019 for NIPA aggregate consumption. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

Panel E of Table 4 reports the GMM estimates using bondholders' consumption. We find that the results are fairly similar to our main results in panel A. For example, when $S = 20$, the estimated risk-aversion coefficient is 14.9 with the cross-sectional $\bar{R}^2$ of 0.80.¹⁹

3.2.2. Tests on the Consistency of Pricing Performance Across Test Assets. In this section, we examine if the long-run risk model prices each set of portfolios consistently with each other. He et al. (2017) emphasize that their intermediary leverage factor produces the estimated price of risk that is consistent across various asset classes. To examine the performance of the long-run risk

model from this perspective, we separately estimate (14) using seven subsamples of portfolios sorted on credit rating, credit spreads, downside risk, idiosyncratic volatility, intermediary betas, long-term reversals, and maturity. Because each subsample has only 5 or 10 portfolios, the risk-aversion parameter will not be as precisely estimated as before. Thus, to evaluate the consistency of estimated γ , we also report the pricing performance in terms of $\bar{R}^2$ and $(RMSE/RMSR)$ for each portfolio group by imposing the same risk aversion estimated by all portfolios.

Panel A of Table 6 shows that risk-aversion estimates are consistent across assets for wealthy households' 20-quarter consumption growth. Once we estimate the

Table 6. GMM Cross-Sectional Regression for Each Portfolio Group

Assets	Credit spread portfolios	Downside portfolios	Maturity portfolios	Rating portfolios	Intermediary portfolios	IdioVol portfolios	Reversal portfolios	All portfolios
Panel A: CEX consumption of wealthy households								
ζ (%)	0.75 [0.33 1.00]	0.64 [−0.01 0.85]	0.20 [−0.46 1.28]	0.82 [0.13 1.17]	0.67 [0.25 1.56]	0.71 [0.10 0.94]	0.81 [0.47 1.13]	0.74 [0.42 0.96]
γ	16.7 [8.7 32.3]	17.1 [6.6 42.6]	35.3 [−14.9 68.6]	14.4 [−2.0 39.0]	14.1 [−19.3 27.6]	15.1 [5.2 40.7]	12.8 [6.6 19.5]	15.4 [7.1 25.9]
$\bar{R}^2$	0.94 [0.36 0.98]	0.96 [0.68 1.00]	0.56 [0.01 0.96]	0.96 [0.06 0.99]	0.24 [0.00 0.87]	0.87 [0.45 0.99]	0.69 [0.30 0.94]	0.80 [0.26 0.90]
$\bar{R}^2$ with same γ	0.93	0.95	0.37	0.96	0.24	0.87	0.66	0.80
$\frac{RMSE}{RMSR}$	0.08	0.06	0.12	0.03	0.15	0.10	0.16	0.12
$\frac{RMSE}{RMSR}$ with same γ	0.09	0.07	0.14	0.05	0.18	0.10	0.17	0.12
Number of assets	10	5	5	5	5	5	5	40
Number of asset-month	3,690	1,845	1,845	1,845	1,785	1,845	1,805	14,660
Panel B: NIPA aggregate consumption								
ζ (%)	0.17 [−1.24 0.98]	0.02 [−0.75 0.97]	0.53 [0.33 1.13]	0.23 [−1.50 1.04]	0.18 [−0.43 1.14]	0.16 [−0.32 1.00]	0.08 [−0.77 1.29]	0.26 [−0.17 1.01]
γ	59.3 [31.4 113.5]	72.0 [29.2 149.2]	22.6 [−11.8 46.4]	41.5 [17.1 98.8]	47.8 [−24.5 108.3]	63.4 [23.6 123.8]	88.9 [−35.2 141.9]	50.1 [17.1 67.9]
$\bar{R}^2$	0.86 [0.53 0.94]	0.97 [0.49 1.00]	0.43 [0.00 0.73]	0.96 [0.73 0.98]	0.98 [0.01 0.98]	0.98 [0.43 0.99]	0.46 [0.00 0.86]	0.64 [0.09 0.79]
$\bar{R}^2$ with same γ	0.84	0.89	−0.27	0.91	0.98	0.94	0.35	0.64
$\frac{RMSE}{RMSR}$	0.15	0.05	0.10	0.05	0.03	0.04	0.22	0.20
$\frac{RMSE}{RMSR}$ with same γ	0.17	0.13	0.23	0.20	0.15	0.10	0.33	0.20
Number of assets	10	5	5	5	5	5	5	40
Number of asset-month	5,300	2,540	2,660	2,660	2,480	2,540	2,380	20,560

Notes. This table reports GMM cross-sectional regression results for each portfolio group: $\hat{E}[r_{i,t+1} - r_{f,t}] + (\hat{\sigma}^2(r_{i,t+1}))/2 - (\hat{\sigma}^2(r_{f,t}))/2 = \zeta + (\gamma - 1)c\hat{o}v(\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}) + e_{it}$, where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative consumption growth over multiple horizons $\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s})$, where $S = 20$ quarters for CEX wealthy household consumption (panel A) and $S = 8$ quarters for NIPA aggregate consumption (panel B). The quantity of risk is jointly estimated with parameters ζ , η , and γ using GMM. Test assets are 40 portfolios including 10 credit spread-sorted portfolios, 5 downside risk-sorted portfolios, 5 maturity-sorted portfolios, 5 credit rating-sorted portfolios, 5 intermediary factor (He et al. 2017) beta-sorted portfolios, 5 idiosyncratic volatility-sorted portfolios, and 5 long-term reversal portfolios. Reported are the intercepts ζ , η , and implied risk-aversion coefficients γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \bar{R}^e_i) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\bar{R}^e_i$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N) \sum_{i=1}^N (E(R_i^e) - \bar{R}^e_i)^2}$ and $RMSR = \sqrt{(1/N) \sum_{i=1}^N E(R_i^e)^2}$. " $\bar{R}^2$ with same γ " and " $(RMSE/RMSR)$ with same γ " report the pricing performance by imposing γ estimated using all portfolios. The time period spans from March 1984 to December 2019 for CEX and from February 1973 to December 2019 for NIPA. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

Euler equation separately, the estimated γ is 17 for credit spread portfolios, 17 for downside risk portfolios, 35 for maturity portfolios, 14 for credit rating portfolios, 14 for intermediary beta portfolios, 15 for idiosyncratic volatility portfolios, and 13 for long-term reversal portfolios. Table 6 also compares $\bar{R}^2$ calculated with the risk aversion specific to each portfolio group with $\bar{R}^2$ with the risk aversion fixed at the value using all portfolios (i.e., $\gamma = 15$). These $\bar{R}^2$ values are similar to each other, showing that allowing the risk-aversion parameter to vary across subsamples does not affect the fit of the model. The cross-sectional $\bar{R}^2$ for the intermediary beta portfolios is low at 0.24, but we will later show that the fit of the model improves significantly if we account for infinite-horizon consumption growth.

Figure 2 displays point estimates of risk aversion from each portfolio group and the one from all portfolio groups, as well as their 95% confidence intervals using the consumption of wealthy households. The two standard error bounds for these estimates include risk aversion of 15 from the main results in Table 4. Therefore, we cannot reject the hypothesis that the coefficient of risk aversion is equal to 15 for each portfolio group. This shows that although the measurement errors for subsamples are large, the estimated risk aversion is consistent across different test assets.²⁰

Figures 3 and 4 show the fit of the model graphically for CEX wealthy households' consumption and NIPA aggregate consumption, respectively. We plot the model-implied prediction for expected returns and the average excess returns in the data using both full samples and seven subsamples of bond portfolios. In the figures, the observations lie close to the 45° line for most subportfolios. A notable exception is the fifth quintile of maturity and intermediary beta-sorted portfolios, which has higher average excess returns than the model's prediction. Because of these somewhat anomalous portfolios, the cross-sectional $\bar{R}^2$ using these sets of portfolios is lower than other portfolios. Still, overall results support the performance of the long-run risk model, which

generates consistent γ with similar magnitudes across different test assets using either wealthy households' consumption or aggregate consumption. Because risk-aversion estimates using wealthy households' consumption are more in line with theories, we focus on the consumption risk of wealthy households for the following analyses.

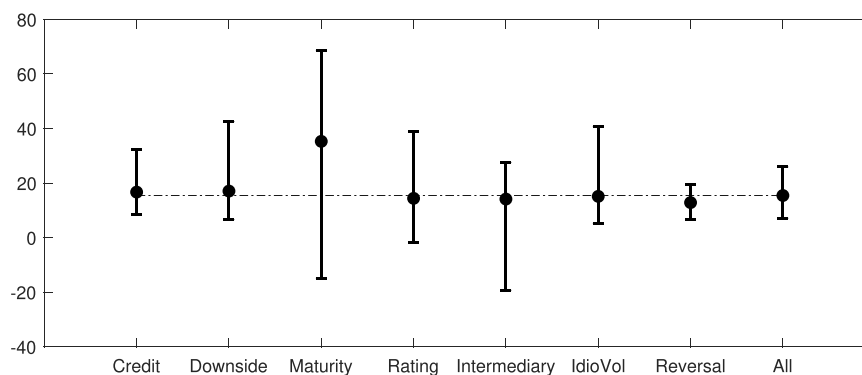
3.3. More General Models Using a VAR

In the previous section, we use a 20-quarter consumption growth for a risk factor as an approximation to the long-run risk model, which in principle, depends on shocks to infinite-horizon consumption expectation. Although the performance of the approximate model is encouraging, we need to ensure that the results are not driven by the simplifying assumption. Thus, in this section, we estimate the conditional model in (11), which requires the estimates of the conditional expectation for long-run consumption growth, $E_t \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s})$. To this end, we choose a vector of state variables and estimate the VAR in (3) and (4). To save space, most of the estimation results are reported in Online Appendix VII.

To begin, we select a candidate set of state variables that predict consumption growth. To avoid an arbitrary choice of a few state variables, we rely on a large set of 160 macro and financial variables listed in Table IA5 in the online appendix.²¹ Then, we apply the principal component analysis as in Ludvigson and Ng (2007, 2009) and Roussanov (2014) and extract 10 factors: $F_1, \dots, F_{10}$. In total, we consider 10 principal components with the maximum VAR lags of two as a candidate set of state variables. Next, based on these candidate variables, we find a subset that minimizes the Akaike information criterion for consumption predictive regression in (3).

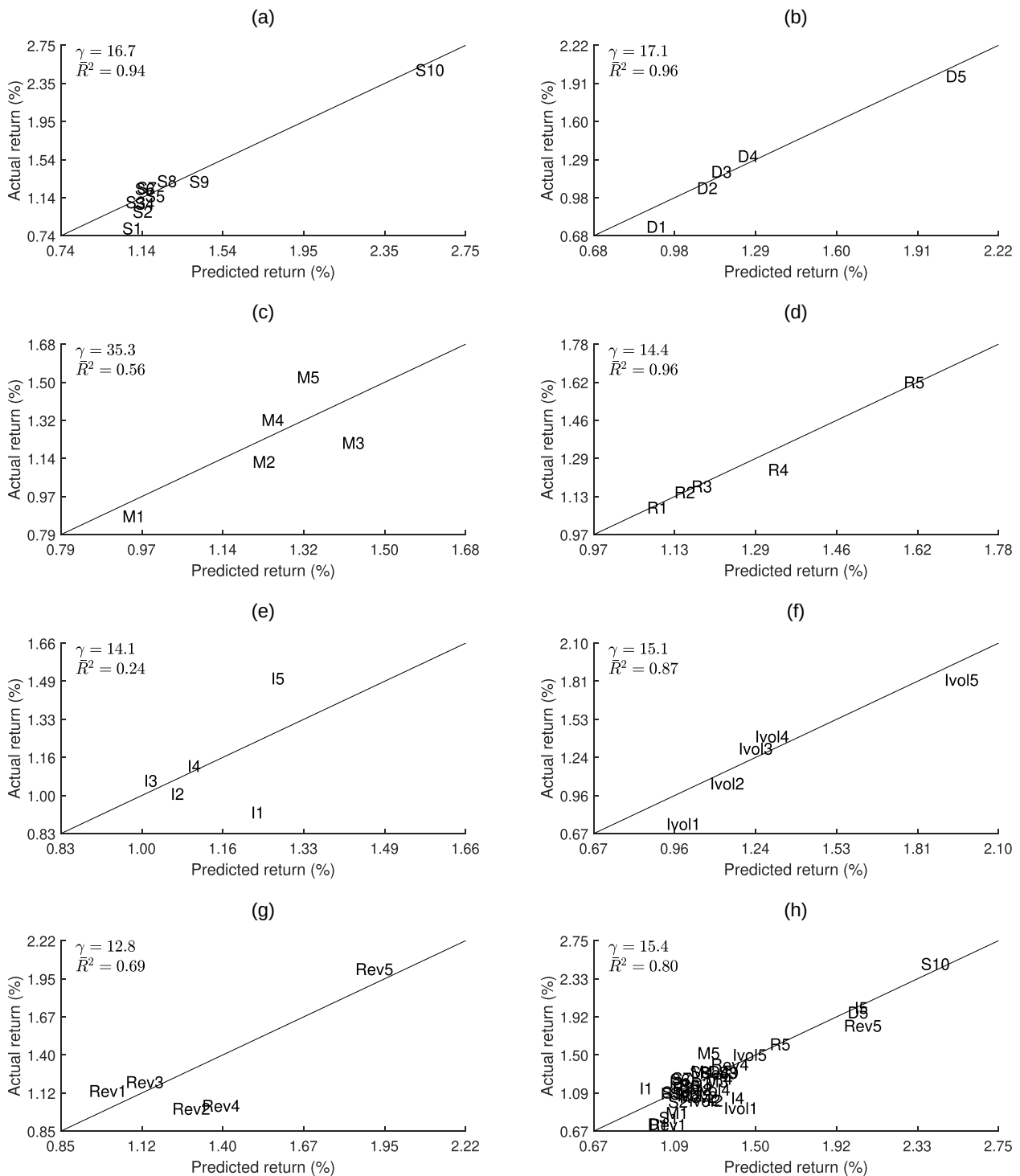
Panel A of Table 7 presents the estimated coefficients of VARs in (3) and (4). We find that the set of state variables we choose predicts consumption growth well. F_2 , F_6 , F_8 , and their one-month lags predict three months ahead of consumption growth with an adjusted R^2 of 0.027. We emphasize that we do not specify VARs based

Figure 2. Risk-Aversion Coefficients by Each Portfolio Group

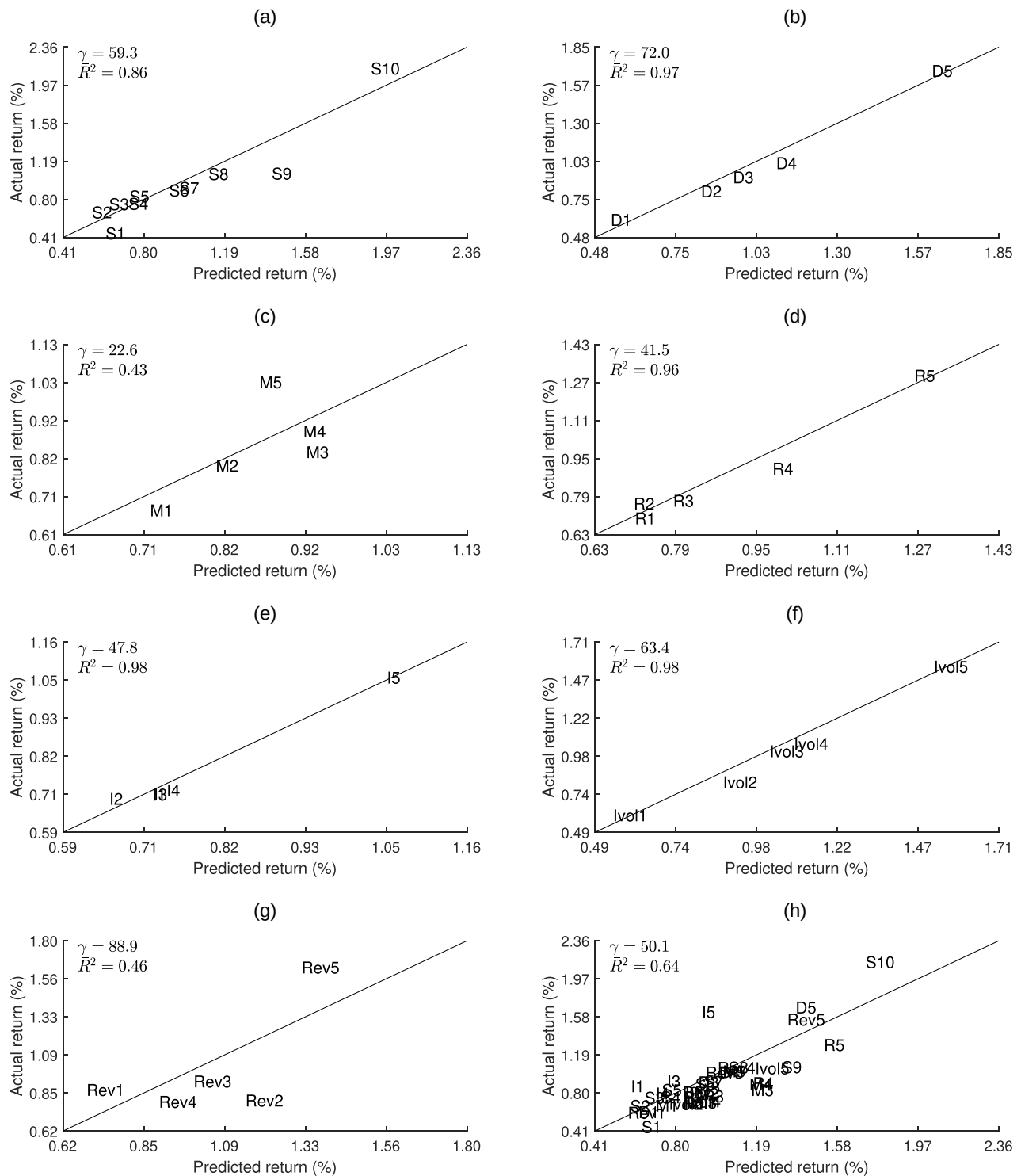


Notes. This figure plots the implied risk-aversion coefficients from GMM cross-sectional regressions for each portfolio group with two standard error bounds. IdioVol refers to 5 idiosyncratic volatility-sorted portfolios.

Figure 3. Fitted and Average Returns of Bonds by Long-Run Risk Using Wealthy Households' Consumption



Notes. This figure plots the cross-section of actual average bond excess returns against the predicted average bond excess returns by the long-run consumption risk using wealthy households' consumption and the GMM estimation. (a) Ten credit spread-sorted portfolios. (b) Five downside-sorted portfolios. (c) Five maturity-sorted portfolios. (d) Five rating-sorted portfolios. (e) Five intermediary beta-sorted portfolios. (f) Five idiosyncratic volatility-sorted portfolios. (g) Five long-term reversal portfolios. (h) All portfolios (40).

Figure 4. Fitted and Average Returns of Bonds by Long-Run Risk Using Aggregate Consumption

Notes. This figure plots the cross-section of average bond excess returns against the predicted average bond excess returns by the long-run consumption risk using NIPA aggregate consumption and the GMM estimation. (a) Ten credit spread-sorted portfolios. (b) Five downside-sorted portfolios. (c) Five maturity-sorted portfolios. (d) Five rating-sorted portfolios. (e) Five intermediary beta-sorted portfolios. (f) Five idiosyncratic volatility-sorted portfolios. (g) Five long-term reversal portfolios. (h) All portfolios (40).

Table 7. VAR Estimation

	x_{t-2}								
Dependent variable	$F_{2,t-2}$	$F_{6,t-2}$	$F_{8,t-2}$	$F_{2,t-3}$	$F_{6,t-3}$	$F_{8,t-3}$	Constant	Adj. R^2	N
Panel A: VAR estimation									
$c_{t+1} - c_{t-2}$	−0.0020 (0.0158)	0.0484 (0.0247)	0.1379 (0.0398)	−0.0294 (0.0151)	0.0280 (0.0251)	−0.0960 (0.0394)	−0.0088 (0.0047)	0.0275	430
$F_{2,t+1}$	0.0197 (0.0568)	0.0511 (0.0821)	−0.1162 (0.1251)	0.0213 (0.0751)	−0.0231 (0.0680)	0.1120 (0.1399)		−0.0200	430
$F_{6,t+1}$	0.0333 (0.0290)	0.1344 (0.0501)	−0.0148 (0.0863)	0.0225 (0.0318)	0.0879 (0.0600)	0.0698 (0.0813)		0.0141	430
$F_{8,t+1}$	0.0096 (0.0203)	0.0123 (0.0259)	0.4828 (0.0439)	−0.0105 (0.0174)	0.0359 (0.0298)	0.3239 (0.0434)		0.2941	430
$F_{2,t}$	0.0076 (0.0538)	0.1205 (0.1112)	0.0942 (0.2185)	0.0183 (0.0786)	0.0313 (0.0767)	−0.1017 (0.2003)		−0.0154	430
$F_{6,t}$	−0.0536 (0.0322)	0.1512 (0.0491)	−0.0613 (0.1013)	0.0641 (0.0273)	0.1322 (0.0530)	0.0708 (0.0780)		0.0439	430
$F_{8,t}$	0.0323 (0.0187)	−0.0021 (0.0284)	0.3922 (0.0419)	−0.0216 (0.0169)	0.0387 (0.0272)	0.4258 (0.0419)		0.3158	430
Panel B: Predictive power for credit spread-sorted decile portfolio returns									
$r_{t-2 \rightarrow t+1}^{S1}$	0.0118 (0.0057)	−0.0093 (0.0063)	−0.0012 (0.0104)	0.0049 (0.0037)	−0.0009 (0.0068)	0.0022 (0.0114)	0.0165 (0.0023)	0.0302	426
$r_{t-2 \rightarrow t+1}^{S2}$	0.0149 (0.0080)	−0.0115 (0.0091)	−0.0069 (0.0130)	0.0093 (0.0052)	0.0037 (0.0097)	−0.0108 (0.0145)	0.0200 (0.0027)	0.0364	426
$r_{t-2 \rightarrow t+1}^{S3}$	0.0117 (0.0083)	−0.0140 (0.0103)	−0.0078 (0.0145)	0.0081 (0.0061)	0.0038 (0.0107)	−0.0072 (0.0161)	0.0208 (0.0029)	0.0177	426
$r_{t-2 \rightarrow t+1}^{S4}$	0.0124 (0.0078)	−0.0108 (0.0109)	−0.0072 (0.0142)	0.0081 (0.0064)	0.0066 (0.0107)	−0.0159 (0.0163)	0.0216 (0.0029)	0.0166	426
$r_{t-2 \rightarrow t+1}^{S5}$	0.0097 (0.0078)	−0.0122 (0.0110)	−0.0165 (0.0159)	0.0111 (0.0065)	0.0036 (0.0109)	−0.0112 (0.0171)	0.0228 (0.0030)	0.0148	426
$r_{t-2 \rightarrow t+1}^{S6}$	0.0071 (0.0070)	−0.0164 (0.0110)	−0.0132 (0.0159)	0.0109 (0.0065)	−0.0012 (0.0113)	−0.0076 (0.0174)	0.0230 (0.0031)	0.0117	426
$r_{t-2 \rightarrow t+1}^{S7}$	0.0035 (0.0063)	−0.0230 (0.0129)	−0.0114 (0.0156)	0.0128 (0.0072)	−0.0047 (0.0116)	−0.0080 (0.0177)	0.0232 (0.0031)	0.0169	426
$r_{t-2 \rightarrow t+1}^{S8}$	−0.0015 (0.0061)	−0.0219 (0.0136)	−0.0228 (0.0167)	0.0105 (0.0071)	−0.0037 (0.0113)	−0.0175 (0.0176)	0.0260 (0.0030)	0.0144	426
$r_{t-2 \rightarrow t+1}^{S9}$	−0.0056 (0.0081)	−0.0308 (0.0162)	−0.0180 (0.0214)	0.0140 (0.0092)	−0.0070 (0.0139)	−0.0191 (0.0219)	0.0259 (0.0033)	0.0171	426
$r_{t-2 \rightarrow t+1}^{S10}$	−0.0411 (0.0202)	−0.0536 (0.0321)	0.0139 (0.0440)	−0.0077 (0.0204)	−0.0269 (0.0285)	−0.0289 (0.0547)	0.0340 (0.0082)	0.0167	426

Notes. Panel A reports the VAR estimation with the consumption growth and the selected state variables. The system of equations is estimated using OLS equation by equation. $F_{n,t}$ is the n th factor from the principal component analysis factors based on 160 preselected variables. Panel B reports the predictive OLS regression of quarterly returns to the credit spread-sorted decile portfolios on the selected state variables. S1 and S10 denote the lowest and highest credit spread portfolios, respectively. Standard errors based on Newey and West (1987) are reported in parentheses. The lag for the standard errors is automatically selected based on Newey and West (1994). The time period spans from March 1984 to December 2019. t denotes a month t .

on how shocks to state variables covary with bond returns. Instead, the state variables are selected based only on how well they predict consumption growth.²²

With this VAR estimate, we follow Hansen et al. (2007, 2008) and derive a more general form of the log stochastic discount factor for the long-run risk model with Epstein–Zin utility, which accounts for the conditional expectation of consumption growth and does not assume EIS = 1. When EIS $\neq$ 1, the factor is no longer a linear function of state variables, and thus, we need to

log linearize the function with approximation. Specifically, without constant terms and a contemporaneous consumption growth ($c_{t+1} - c_t$) term that does not materially affect our result, the first-order expansion of the logarithm of the stochastic discount factor yields

$$s_{t+1} \approx (1 - \gamma)\lambda(\delta)w_{t+1} + \left(\frac{1}{\rho} - 1\right) \times \left(\frac{1}{2}w'_{t+1}\Theta_0w_{t+1} + w'_{t+1}\Theta_1x_t + \theta_1x_t + \theta_2w_{t+1}\right), \quad (16)$$

Table 8. Tests Using the Long-Run Risk Measure Based on VAR

Assets	Credit spread portfolios	Downside portfolios	Maturity portfolios	Rating portfolios	Intermediary portfolios	IdioVol portfolios	Reversal portfolios	All portfolios
ζ (%)	0.75 [0.41 1.01]	0.51 [0.09 0.80]	0.67 [0.21 1.36]	0.89 [0.60 1.16]	0.60 [0.36 1.16]	0.69 [0.13 0.92]	0.64 [0.27 0.96]	0.74 [0.43 1.12]
γ	19.6 [8.7 43.7]	25.2 [7.9 62.3]	24.5 [−23.9 68.9]	14.0 [−1.8 37.1]	30.6 [−17.1 43.9]	18.9 [5.3 63.7]	26.5 [13.3 40.4]	19.7 [4.0 41.8]
$\bar{R}^2$	0.96 [0.49 0.98]	0.99 [0.56 0.99]	0.26 [0.00 0.92]	0.88 [0.04 0.97]	0.90 [0.01 0.97]	0.92 [0.49 0.99]	0.64 [0.20 0.92]	0.83 [0.07 0.89]
$\bar{R}^2$ with same γ	0.96	0.94	0.25	0.71	0.78	0.91	0.60	0.83
$\frac{RMSE}{RMSR}$	0.06	0.03	0.16	0.05	0.05	0.09	0.16	0.11
$\frac{RMSE}{RMSR}$ with same γ	0.06	0.10	0.16	0.08	0.09	0.11	0.17	0.11
Number of assets	10	5	5	5	5	5	5	40
Number of asset-month	4,260	2,130	2,130	2,130	2,070	2,130	2,090	16,940

Notes. This table presents GMM cross-sectional test results using the long-run risk measure based on VAR. The long-run consumption risk factor is measured as $(\hat{E}_{t+1} - \hat{E}_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s})$. The quantity of risk is jointly estimated with parameters ζ and γ using GMM. Consumption of wealthy households defined as the top 30% of asset holders from CEX data is used. Test assets are 10 credit spread-sorted portfolios, 5 downside risk-sorted portfolios, 5 maturity-sorted portfolios, 5 credit rating-sorted portfolios, 5 intermediary factor (He et al. 2017) beta-sorted portfolios, 5 idiosyncratic volatility-sorted portfolios, and 5 long-term reversal portfolios; 95% confidence intervals for parameters based on bootstrapping with 5,000 replications are reported in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \hat{R}^e_i) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\hat{R}^e_i$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N) \sum_{i=1}^N (E(R_i^e) - \hat{R}^e_i)^2}$ and $RMSR = \sqrt{(1/N) \sum_{i=1}^N E(R_i^e)^2}$. “ $\bar{R}^2$ with same γ ” and “ $(RMSE/RMSR)$ with same γ ” report the pricing performance by imposing γ estimated using all portfolios. The time period spans from March 1984 to December 2019. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

where $\lambda(\delta)w_{t+1} = (E_{t+1} - E_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s})$ and Θ_0, Θ_1 , and Θ_2 are functions of the parameters for the dynamics of state variables described in Online Appendix VII. The first term in (16) represents the log SDF when an EIS parameter $\rho = 1$, whereas the second term arises when $\text{EIS} \neq 1$. When EIS is one, the model does not require any restrictions on the shock structure to identify w_{t+1} . For this reason, we first focus our VAR analysis on the case for $\text{EIS} = 1$. We then conduct the analysis for the general case where $\text{EIS} \neq 1$ by identifying w_{t+1} .

With the assumption of $\text{EIS} = 1$, we calculate shocks to the long-run expectation

$$(E_{t+1} - E_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s}) = \epsilon_{c,t+1} + \delta U_c (I - \delta G)^{-1} \epsilon_{x,t+1}, \quad (17)$$

where $\epsilon_{c,t+1} = \eta_0 w_{t+1}$ and $\epsilon_{x,t+1} = H w_{t+1}$. We use regression residuals of (3) and (4) for $\epsilon_{c,t+1}$ and $\epsilon_{x,t+1}$. The standard deviation of VAR-based shocks to long-run consumption is 8.19% per quarter, which is slightly lower than the standard deviation of unconditional 20-quarter consumption growth (8.82% per quarter).²³

Armed with the estimated shocks to expectation for the long-run consumption growth in (17), we estimate the conditional model in (11) using GMM. Table 8 reports the estimates. Using all 40 portfolios as test assets, we find that the estimated risk-aversion parameter γ is 20. Although this point estimate is slightly higher than

the estimate for the unconditional model of 15, a coefficient of 10 is well within two standard errors. The cross-sectional $\bar{R}^2$ is 0.83, even higher than 0.80 using the discounted 20-quarter consumption growth as a risk factor.

The analysis of seven subsamples also yields findings similar to our main results in Table 6. The risk-aversion estimates are consistent across portfolio groups with the two standard error bounds that include the full-sample risk-aversion estimate of 20. Moreover, the cross-sectional $\bar{R}^2$ statistics are sizable, ranging from 0.64 to 0.99, except for maturity portfolios. Importantly, the model explains 90% of variations in average returns associated with intermediary factor betas.

Next, we allow EIS to be different from one and estimate the second term of the log SDF in (16). To this end, we impose a structure on the shock vector and estimate parameters in the second term. Online Appendix VII details terms in (16) and describes how we structurally identify the shock vector w_{t+1} . Table 9 presents the GMM cross-sectional test results with different values of EIS from 0.3 to 2.0. For an EIS of 1.5, which is used in Bansal and Yaron (2004), estimated γ is 21, very close to $\gamma = 20$ when $\text{EIS} = 1$, and the cross-sectional $\bar{R}^2$ of 0.84 is almost the same as the one with $\text{EIS} = 1$. The minimum and maximum estimated levels of risk aversion are 15 and 22 when $\text{EIS} = 0.3$ and $\text{EIS} = 2$, respectively. The two standard error bounds include risk-aversion levels around 10 for all values of EIS. Moreover, all risk-aversion estimates are statistically different from zero, and the cross-sectional $\bar{R}^2$ statistics remain sizable regardless of

Table 9. Tests Using the Long-Run Risk Measure Based on VAR with Different EIS Coefficients

Assets	EIS = 0.3	EIS = 0.5	EIS = 0.7	EIS = 1	EIS = 1.5	EIS = 2
ζ (%)	0.93 [0.66 1.29]	0.85 [0.60 1.20]	0.80 [0.54 1.15]	0.74 [0.43 1.12]	0.69 [0.36 1.10]	0.66 [0.34 1.10]
γ	14.6 [4.6 24.3]	16.8 [4.4 31.2]	18.2 [4.2 36.7]	19.7 [4.0 41.8]	21.1 [3.8 41.4]	21.9 [3.8 40.3]
$\bar{R}^2$	0.78 [0.07 0.86]	0.81 [0.07 0.87]	0.82 [0.07 0.88]	0.83 [0.07 0.89]	0.84 [0.07 0.89]	0.84 [0.07 0.89]
$\frac{RMSE}{RMSR}$	0.13	0.12	0.12	0.11	0.11	0.11
Number of assets	40	40	40	40	40	40
Number of asset-month	16,940	16,940	16,940	16,940	16,940	16,940

Notes. This table presents the GMM cross-sectional test results using the long-run risk measure based on VAR with different EIS coefficients. To allow EIS to be different from one, we estimate the following log SDF: $s_{t+1} \approx (1 - \gamma)\lambda(\delta)w_{t+1} + (1/\rho - 1) \left((1/2)w'_{t+1}\Theta_0w_{t+1} + w'_{t+1}\Theta_1x_t + \Theta_1x_t + \Theta_2w_{t+1} \right)$. The quantity of risk is jointly estimated with parameters of ζ and γ using GMM. Consumption of wealthy households defined as the top 30% of asset holders from CEX data is used. Test assets are 10 credit spread-sorted portfolios, 5 downside risk-sorted portfolios, 5 maturity-sorted portfolios, 5 credit rating-sorted portfolios, 5 intermediary factor (He et al. 2017) beta-sorted portfolios, 5 idiosyncratic volatility-sorted portfolios, and 5 long-term reversal portfolios. Reported are the intercepts ζ and implied risk-aversion coefficients γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \hat{R}_i^e) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\hat{R}_i^e$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N)\sum_{i=1}^N (E(R_i^e) - \hat{R}_i^e)^2}$ and $RMSR = \sqrt{(1/N)\sum_{i=1}^N E(R_i^e)^2}$. The time period spans from March 1984 to December 2019. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

the values of EIS. These results suggest that a key driver of the stochastic discount factor in (16) is its first term as used in our main exercise, and the second term that is added in this extension plays a minor role in explaining the cross-section of corporate bond risk premiums. The fact that the results are not sensitive to the choice of EIS justifies our initial assumption of the unitary EIS in the main results. In Online Appendix VII, we extend the VAR to account for a volatility shock to the SDF as in Bansal et al. (2014) and confirm that the results are similar.

In sum, we find that either unconditional or conditional measures of the long-run risk explain the corporate bond risk premiums well.

4. Extension

4.1. Departure from the Lognormality Assumption

In deriving the Euler equation for log excess returns in (9), we assume that consumption growth and asset returns are jointly lognormally distributed. Although this is a standard assumption in the literature, it may not be suitable for our sample of corporate bonds. Corporate bond returns may not be lognormally distributed because in the Merton (1974) model, corporate bonds are viewed as a portfolio of risk-free assets and a short position on a put option on the underlying firm's asset, and thus, their payoff is a nonlinear function of the asset value. This feature of corporate bonds' payoff may lead to the skewed distribution of bond returns, invalidating the Euler equation derived in (9).

Harvey and Siddique (2000) present an empirical framework to test an asset pricing model accounting for skewness in asset returns. To account for the dependence in higher-order moments between factors and

asset returns, Harvey and Siddique (2000) express expected returns as a function of covariance and coskewness between factors and asset returns. We follow their approach and expand the Euler equation up to the third order:

$$1 = E[S_{t+1}R_{i,t+1}], \approx \bar{G} \left(1 + E[r_{i,t+1}] + \frac{1}{2}E[r_{i,t+1}^2] + (1 - \gamma)E[\varepsilon_{c,t \rightarrow t+S}r_{i,t+1}] \right) \quad (18)$$

$$+ \frac{1}{2}(1 - \gamma)^2E[\varepsilon_{c,t \rightarrow t+S}^2r_{i,t+1}] + \frac{1}{2}(1 - \gamma)E[\varepsilon_{c,t \rightarrow t+S}r_{i,t+1}^2] + \text{Const} \Big), \quad (19)$$

where $\bar{G} = e^{\log \delta - 0.5(1-\gamma)^2\eta(\delta)^2 + (1-\gamma)\mu_{\sum_{s=0}^{\infty} \delta^s \Delta c_{t+1+s}}}$. Equation (18) shows that the risk premiums depend on covariance as well as coskewness.²⁴ Thus, we estimate the GMM using the moment conditions:

$$0 = E \left[\begin{pmatrix} r_{i,t+1} - r_{f,t} + \frac{E(r_{i,t+1}^2)}{2} - \frac{E(r_{f,t}^2)}{2} - \zeta - (\gamma - 1) \\ \times \varepsilon_{c,t \rightarrow t+S}(r_{i,t+1} - r_{f,t}) \\ + \frac{1}{2}(1 - \gamma)^2\varepsilon_{c,t \rightarrow t+S}^2(r_{i,t+1} - r_{f,t}) - \frac{1}{2}(\gamma - 1) \\ \times \varepsilon_{c,t \rightarrow t+S}(r_{i,t+1}^2 - r_{f,t}^2) \\ \varepsilon_{c,t \rightarrow t+S} \end{pmatrix} \right]. \quad (20)$$

Unlike Harvey and Siddique (2000), the regression specification in (20) restricts the set of free parameters, (ζ, γ) , to match those in the main analysis because the loadings on the covariance and coskewness both depend on the risk-aversion parameter γ .

Table 10. Test Accounting for Skewness

	Value
ζ (%)	1.25 [0.17 0.86]
γ	8.1 [3.1 9.7]
$\bar{R}^2$	0.38 [0.05 0.55]
$\frac{RMSE}{RMSR}$	0.21
Number of assets	40
Number of asset-month	14,660

Notes. This table reports a GMM cross-sectional regression result that accounts for skewness of bond returns: $\hat{E}[r_{i,t+1} - r_{f,t+1}] + (1/2)\hat{E}[r_{i,t+1}^2] - (1/2)\hat{E}[r_{f,t+1}^2] = \zeta + (\gamma - 1)\hat{E}[\epsilon_{c,t \rightarrow t+S}(r_{i,t+1} - r_{f,t+1})] - (1/2)(1 - \gamma)^2\hat{E}[\epsilon_{c,t \rightarrow t+S}^2(r_{i,t+1} - r_{f,t+1})] + (1/2)(\gamma - 1)\hat{E}[\epsilon_{c,t \rightarrow t+S}(r_{i,t+1}^2 - r_{f,t+1}^2)] + e_{i,t}$, where $\epsilon_{c,t \rightarrow t+S} = \sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s}) - \mu_{\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s})}$, $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative 20-quarter consumption growth ($S = 20$). Consumption of wealthy households defined as the top 30% of asset holders from CEX data is used. The quantity of risk is jointly estimated with parameters ζ and γ . Test assets are 40 portfolios including 10 credit spread-sorted portfolios, 5 downside risk-sorted portfolios, 5 maturity-sorted portfolios, 5 credit rating-sorted portfolios, 5 intermediary factor (He et al. 2017) beta-sorted portfolios, 5 idiosyncratic volatility-sorted portfolios, and 5 long-term reversal portfolios. Reported are the intercepts ζ and implied risk-aversion coefficient γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \hat{R}_i^e) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\hat{R}_i^e$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N) \sum_{i=1}^N (E(R_i^e) - \hat{R}_i^e)^2}$ and $RMSR = \sqrt{(1/N) \sum_{i=1}^N E(R_i^e)^2}$. The time period spans from March 1984 to December 2019. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

Table 10 presents the estimated slope coefficients in (20) and the fit of the model using 40 bond portfolios as test assets. We find that the estimated γ is 8, much lower than γ of 15 from our main result. In terms of the cross-sectional fit, $\bar{R}^2$ is 0.38, also lower than the main result. Overall, we find evidence for the improvement of the level of risk aversion as we account for skewness in bond returns, although the pricing performance is not as strong as before in this case.

4.2. International Corporate Bonds

As another extension, we perform an out-of-sample test using U.S. dollar-denominated corporate bonds issued by foreign firms. Specifically, we use bonds in the ICE Bank of America Merrill Lynch Global Corporate Index and High Yield Index. We focus on 30 economies with a nontrivial number of observations every year, including Australia, Belgium, Brazil, Canada, Chile, China, Colombia, France, Germany, Hong Kong, India, Israel, Italy, Japan, Malaysia, Mexico, the Netherlands, Norway, Peru,

Qatar, Russia, Singapore, South Korea, Spain, Sweden, Switzerland, Thailand, Turkey, the United Arab Emirates, and the United Kingdom. We use U.S. dollar-denominated bonds because according to Maggiori et al. (2020), there is a home currency bias for bond investors, and thus, those bonds are likely to be included in the U.S. investors' portfolios. Using those bonds, we form 40 portfolios based on the seven characteristics as in the main results. We then estimate the Euler equation in (14) and report the estimated risk-aversion coefficients in Table 11.

Using the international corporate bond portfolios, the estimated risk-aversion coefficient is 11 for the 20-quarter consumption growth of wealthy households, which is close to the estimate of 15 in the main results. Because the model generates a reasonable cross-sectional R^2 of 0.46, long-run consumption risk explains these international bonds with a parameter similar to the main results for the U.S. corporate bonds.

4.3. Time Series Analysis Using Factor-Mimicking Portfolios and Comparison with Other Factor Models

In this section, we compare the performance of our consumption risk factor with that of other risk factors proposed in the literature. To this end, we follow the econometric framework of Barillas and Shanken (2018) and regress a risk factor on other factors to study whether it carries risk premiums unexplained by the other factors. Because this approach requires the factor to be an excess return, we create a factor-mimicking portfolio for the wealthy households' long-run consumption risk. To construct the mimicking portfolio, we project the discounted 20-quarter cumulative consumption growth (unconditional measure) or shocks to the long-run expectation (conditional measure) on quarterly excess returns of six bond portfolios independently sorted on three maturity bins and two credit rating bins (investment grade and high yield),

$$(\text{Long-run consumption shock})_{t+1} = a + b' R_{t+1} + u_{t+1}, \quad (21)$$

where $(\text{Long-run consumption shock})_{t+1} = \sum_{s=0}^{19} \delta^s(c_{t+1+s} - c_{t+s})$ or $(\hat{E}_{t+1} - \hat{E}_t) \sum_{s=0}^{\infty} \delta^s(c_{t+1+s} - c_{t+s})$. A factor-mimicking portfolio is given by the fitted value of this projection $\hat{b}' R_{t+1}$. To extend the sample period, we estimate b using the sample after March 1984 and assume that the same b applies to the earlier sample from April 1973, resulting in the sample extended by roughly 10 years. We find that the average excess returns on the factor-mimicking portfolio are 0.68% and 0.51% per quarter for the unconditional and conditional measures, respectively.

For comparison, we gather 10 asset pricing factors proposed in the literature: the value, size, profitability, investment, term, and default factors of Fama and French

Table 11. Test Using International Corporate Bonds

S (quarters) =	1	2	4	8	12	16	20	24
ζ (%)	0.89 [0.33 1.51]	0.68 [0.28 1.19]	0.89 [0.41 1.31]	1.13 [0.46 1.49]	1.16 [0.46 1.56]	0.95 [0.48 1.43]	0.92 [0.41 1.29]	1.15 [0.47 1.69]
γ	32.1 [−13.0 58.8]	15.0 [−0.4 37.5]	18.4 [−1.5 29.0]	32.2 [−27.3 56.8]	−0.9 [−26.9 43.9]	8.0 [−3.4 30.0]	11.0 [−3.3 28.4]	13.9 [−17.3 40.8]
$\bar{R}^2$	0.29 [0.00 0.83]	0.71 [0.02 0.92]	0.49 [0.01 0.84]	0.23 [0.00 0.83]	0.00 [0.00 0.78]	0.29 [0.00 0.74]	0.46 [0.01 0.90]	0.15 [0.00 0.72]
$\frac{RMSE}{RMSR}$	0.24	0.15	0.20	0.25	0.28	0.20	0.19	0.24
Number of assets	40	40	40	40	40	40	40	40
Number of asset-month	9,435	9,435	9,435	9,435	9,075	8,595	8,115	7,635

Notes. This table reports GMM cross-sectional regression results using 40 characteristics-sorted portfolios based on 30 countries with different long-run horizons S : $\hat{E}[r_{i,t+1} - r_{f,t}] + (\sigma^2(r_{i,t+1}))/2 - (\sigma^2(r_{f,t}))/2 = \zeta + (\gamma - 1)c\hat{ov}(\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}) + e_i$, where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative consumption growth over multiple horizons $\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s})$. The quantity of risk is jointly estimated with parameters ζ and γ using GMM. Consumption of wealthy households defined as the top 30% of asset holders from CEX data is used. Test assets are 40 portfolios including 10 credit spread-sorted portfolios, 5 downside risk-sorted portfolios, 5 maturity-sorted portfolios, 5 credit rating-sorted portfolios, 5 intermediary factor (He et al. 2017) beta-sorted portfolios, 5 idiosyncratic volatility-sorted portfolios, and 5 long-term reversal portfolios. The 30 countries are Australia, Belgium, Brazil, Canada, Chile, China, Colombia, France, Germany, Hong Kong, India, Israel, Italy, Japan, Malaysia, Mexico, the Netherlands, Norway, Peru, Qatar, Russia, Singapore, South Korea, Spain, Sweden, Switzerland, Thailand, Turkey, United Arab Emirates, and the United Kingdom. Reported are the intercepts ζ and implied risk-aversion coefficients γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \hat{R}_i^e) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\hat{R}_i^e$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}^2$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N) \sum_{i=1}^N (E(R_i^e) - \hat{R}_i^e)^2}$ and $RMSR = \sqrt{(1/N) \sum_{i=1}^N E(R_i^e)^2}$. The time period spans from February 1997 to December 2017. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

(1993, 2015); the intermediary capital factor of He et al. (2017); and the bond market, downside risk, and credit risk factors of Bai et al. (2019). Because the intermediary capital factor is not an excess return, we project it on the same base assets as we did for the consumption risk factor using (21) and create an intermediary factor-mimicking portfolio.

Panel A1 of Table 12 reports the average excess returns on the other factors and the estimates for the regression of these factors on the consumption-mimicking portfolio returns. We find that, except for the size and default factors, the factors proposed in the literature earn significantly positive risk premiums during our sample period.

The intercept of the regressions measures the risk premiums on these factors unexplained by the consumption-mimicking portfolio. The estimates in panel A1 suggest that the bond risk factors have alphas lower than the average excess returns. For example, the bond market, downside risk, credit risk, and intermediary risk factors have alphas of -0.08% , -0.12% , -0.06% , and -0.39% , respectively, and these estimates are insignificantly different from zero. In contrast, we find that the equity risk factors are largely orthogonal to the consumption-mimicking portfolio with near-zero R^2 . As a result, the estimated alphas are similar to the average excess returns. These results are expected because the consumption-mimicking portfolio is created by a projection on bond returns.

In panel A2, we repeat the exercise using the mimicking portfolios for the VAR-based long-run consumption shocks. This alternative mimicking portfolio generates

higher betas for the other bond factors, lowering their alphas even further. Overall, we find that the long-run consumption factor explains other bond factors well.

Next, we switch the left-hand side and right-hand side of the regression and regress the consumption-mimicking portfolios on each of the other risk factors. Panels B1 and B2 of Table 12 report the estimated intercepts, slope coefficients, and adjusted R^2 values of the regression.

We find that the alphas of the consumption-mimicking portfolio are positive for all regression specifications. When it is regressed on other bond factors, the alpha is lower than the average excess returns but remains significantly positive. For example, the alphas of the consumption-mimicking portfolio against the bond market, downside risk, credit risk, and intermediary factors are 0.36% , 0.51% , 0.58% , and 0.35% , respectively. When regressed on the stock factors, the alphas are virtually the same as the average excess returns of 0.68% . The results using the VAR-based consumption risk, reported in panel B2, are highly similar to the one using the 20-quarter consumption growth. These results show that the consumption risk factor is not subsumed by other bond and equity risk factors. Furthermore, it carries information on corporate bond risk premiums that explain the existing bond factors.

4.4. Illiquidity of Corporate Bonds

Bai et al. (2019) show that illiquidity measures predict the cross-section of corporate bond returns. To examine whether the consumption-based model explains the

Table 12. Time Series Regressions Using Other Factors and the Long-Run Risk Factor

Factors	<i>HKM</i>	<i>SMB</i>	<i>HML</i>	<i>DEF</i>	<i>TERM</i>	<i>RMW</i>	<i>CMA</i>	<i>BOND-MKT</i>	<i>DOWNSIDE</i>	<i>CREDIT</i>
Panel A: Regression of other factors on the consumption-mimicking portfolio returns										
Panel A1: CEX unconditional long-run risk measure										
Avg. Returns (%)	1.79 (0.33)	0.50 (0.33)	0.97 (0.28)	0.03 (0.09)	1.00 (0.25)	0.84 (0.25)	0.94 (0.19)	0.87 (0.20)	1.10 (0.28)	0.57 (0.18)
α (%)	−0.39 (0.26)	0.25 (0.36)	0.81 (0.29)	−0.32 (0.22)	0.43 (0.48)	1.08 (0.24)	0.97 (0.22)	−0.08 (0.19)	−0.12 (0.33)	−0.06 (0.38)
β	3.22 (0.36)	0.38 (0.15)	0.24 (0.09)	0.52 (0.26)	0.84 (0.46)	−0.35 (0.11)	−0.05 (0.11)	1.40 (0.15)	1.72 (0.42)	0.93 (0.44)
Adj. R^2	0.59	0.02	0.00	0.16	0.08	0.02	0.00	0.52	0.30	0.16
Panel A2: CEX conditional long-run risk measure										
Avg. Returns (%)	1.74 (0.36)	0.50 (0.35)	0.97 (0.30)	0.03 (0.09)	1.00 (0.27)	0.84 (0.27)	0.94 (0.21)	0.87 (0.21)	1.10 (0.30)	0.57 (0.21)
α (%)	−0.96 (0.21)	−0.14 (0.36)	0.84 (0.35)	−0.48 (0.21)	0.83 (0.62)	1.33 (0.35)	1.02 (0.24)	−0.10 (0.25)	−0.81 (0.40)	−0.66 (0.42)
β	5.30 (0.40)	1.26 (0.13)	0.26 (0.19)	0.99 (0.34)	0.33 (0.88)	−0.95 (0.22)	−0.15 (0.12)	1.89 (0.28)	3.55 (0.61)	2.42 (0.65)
Adj. R^2	0.69	0.08	0.00	0.25	0.00	0.07	0.00	0.39	0.57	0.44
Panel B: Regression of the consumption-mimicking portfolio returns on each of other factors										
Panel B1: CEX unconditional long-run risk measure										
α (%)	0.35 (0.11)	0.65 (0.11)	0.65 (0.10)	0.67 (0.11)	0.57 (0.11)	0.73 (0.09)	0.69 (0.11)	0.36 (0.07)	0.51 (0.14)	0.58 (0.12)
β	0.18 (0.02)	0.05 (0.03)	0.03 (0.01)	0.32 (0.06)	0.10 (0.06)	−0.06 (0.06)	−0.01 (0.02)	0.37 (0.03)	0.18 (0.06)	0.17 (0.04)
Adj. R^2	0.59	0.02	0.00	0.16	0.08	0.02	0.00	0.52	0.30	0.16
Panel B2: CEX conditional long-run risk measure										
α (%)	0.28 (0.03)	0.48 (0.06)	0.50 (0.05)	0.50 (0.06)	0.49 (0.07)	0.57 (0.06)	0.52 (0.06)	0.33 (0.05)	0.36 (0.05)	0.41 (0.05)
β	0.13 (0.01)	0.07 (0.02)	0.01 (0.01)	0.25 (0.05)	0.02 (0.04)	−0.07 (0.03)	−0.02 (0.01)	0.21 (0.04)	0.16 (0.02)	0.18 (0.02)
Adj. R^2	0.69	0.08	0.00	0.25	0.00	0.07	0.00	0.39	0.57	0.44

Notes. This table reports time series regressions of other factors on the tradable long-run risk factor in panel A for each factor i : $F_{i,t+1} = \alpha + \beta X_{t+1} + \epsilon_{t+1}$. We also regress the tradable long-run risk factor on other factors in panel B for each factor i : $X_{t+1} = \alpha + \beta F_{i,t+1} + \epsilon_{t+1}$. To estimate the tradable long-run risk factor X_{t+1} , long-run consumption risk measures are regressed on excess returns of six bond portfolios sorted on maturity and credit rating over the CEX sample period (March 1984 to December 2019), and then, the coefficients are used to construct tradable long-run risk factors for April 1973 and December 2019. Panels A1 and B1 report results using the unconditional long-run risk measure, which is the discounted cumulative consumption growth over 20-quarter horizons $\sum_{s=0}^{19} \delta^s (c_{t+1+s} - c_{t+s})$. Panels A2 and B2 report results using the conditional long-run risk measure $(\hat{E}_{t+1} - \hat{E}_t) \sum_{s=0}^{\infty} \delta^s (c_{t+1+s} - c_{t+s})$ based on VAR. Factors that we consider are the intermediary capital risk factor by He et al. (2017), Fama–French seven factors (five factors from Fama and French 2015 in addition to DEF and TERM), aggregate corporate bond excess returns, downside, and credit rating factors. Reported are average returns of each tradable factor, the intercepts α , and coefficient β with bootstrapped standard errors computed with 5,000 replications. Average returns of factors and α are multiplied by 100 for ease of exposition.

illiquidity premium, we sort corporate bonds into quintiles based on the Roll measure (square root of negative autocovariance of daily log price changes calculated in each month), age (time elapsed since issuance), and issue amount of bonds as well as the betas with respect to the noise measure of treasury yield curve proposed by Hu et al. (2013). The Roll measure is available only after July 2002 when TRACE data start. For the noise betas, we compute 36-month rolling betas using the first difference in the noise measure.

We estimate the GMM cross-sectional regressions using 20 illiquidity-sorted portfolios with the CEX wealthy household consumption and NIPA aggregate consumption. In Table 13, with the CEX consumption

of wealthy households, the risk-aversion estimate is statistically indistinguishable from zero. A poor performance of the model in explaining illiquidity-sorted portfolios is exacerbated with the NIPA aggregate consumption, as the estimated γ is negative at -27 in this case. Therefore, the long-run risk model is not a panacea; it explains the cross-section of bond returns likely to be associated with default risk and macroeconomic uncertainty, but it does not explain illiquidity premiums.

5. Conclusion

In this article, we show that a one-factor model based on long-run consumption risk can explain the cross-section of corporate bond returns. Consistent with the literature

Table 13. Test Using Illiquidity Portfolios as Test Assets

Data	CEX	NIPA
ζ (%)	0.46 [0.21 1.60]	1.29 [0.47 1.48]
γ	27.9 [−8.3 38.5]	−27.0 [−54.9 83.1]
$\bar{R}^2$	0.82 [0.02 0.92]	0.33 [0.00 0.78]
$\frac{RMSE}{RMSR}$	0.12	0.29
Number of assets	20	20
Number of asset-month	5,940	7,930

Notes. This table reports GMM cross-sectional regression results using illiquidity portfolios as test assets: $\hat{E}[r_{i,t+1} - r_{f,t}] + (\hat{\sigma}^2(r_{i,t+1})/2 - (\hat{\sigma}^2(r_{f,t}))/2) = \zeta + (\gamma - 1)\hat{c}\hat{v}(\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s}), r_{i,t+1} - r_{f,t}) + e_i$, where $r_{i,t+1}$ is the quarterly log return of an asset i , $r_{f,t}$ is the quarterly log rate of the 30-day T-bill, $\delta = 0.95^{1/4}$, and c_t is the log consumption. The long-run consumption risk factor is measured by the discounted cumulative consumption growth $\sum_{s=0}^{S-1} \delta^s(c_{t+1+s} - c_{t+s})$, where $S = 20$ quarters for CEX and $S = 8$ quarters for NIPA. The column “CEX” reports the result using the consumption growth of wealthy households defined as the top 30% of asset holders from CEX data. The column “NIPA” reports using the consumption growth of aggregate households from NIPA. The quantity of risk is jointly estimated with parameters ζ and γ using GMM. Test assets are five Roll measures of illiquidity-sorted portfolios, five age-sorted portfolios, five issue amount-sorted portfolios, and five noise factor beta-sorted portfolios. Reported are the intercepts ζ and implied risk-aversion coefficients γ with 95% confidence intervals for parameters based on bootstrapping with 5,000 replications in square brackets. The cross-sectional $\bar{R}^2$ is defined as $1 - \text{var}_c(E(R_i^e) - \bar{R}_i^e) / \text{var}_c(E(R_i^e))$, where i is a test asset and $\bar{R}_i^e$ is the predicted average excess return of portfolio i ; 95% confidence intervals for $\bar{R}_i^e$ are reported in square brackets. The pricing error is measured by $(RMSE/RMSR)$, where $RMSE = \sqrt{(1/N) \sum_{i=1}^N (E(R_i^e) - \bar{R}_i^e)^2}$ and $RMSR = \sqrt{(1/N) \sum_{i=1}^N E(R_i^e)^2}$. The time period spans from March 1984 to December 2019 for CEX and from February 1973 to December 2019 for NIPA. Unconditional pricing errors ζ are multiplied by 100 for ease of exposition.

on equity risk premiums, we find that our model explains the most cross-sectional variation in risk premiums associated with credit spreads, maturity, credit rating, downside risk, idiosyncratic volatility, the intermediary factor betas, and long-term reversals with a reasonable risk-aversion coefficient of 15. The performance of the one-factor model suggests that it is possible for a single macroeconomic factor to summarize the various dimensions of corporate bond risk premiums previously described by multifactor models in the literature.

Our finding resonates with the literature that explains credit spreads at the aggregate level using long-run risk models (e.g., Bhamra et al. 2010a, b; Chen 2010; Elkamhi and Salerno 2010). However, in this paper, we directly estimate the quantity and price of risk using bond return data, avoiding the issue of calibrating the model to match poorly estimated historical default frequency.

Our results also point to the reinterpretation of the well-known class of factor models based on shocks to the financial intermediary’s capital (e.g., Adrian et al. 2014, He et al. 2017). The fact that consumption shocks

generate a pattern in covariance that matches average returns is striking given that corporate bonds are mainly owned by financial institutions and traded in the dealer-driven over-the-counter market. However, our findings are not mutually exclusive with the financial intermediary-based explanation of risk premiums. It is possible to argue that depleted intermediary capital causes wealthy households’ expectations for the long-run consumption growth to fall or vice versa. As pointed out by Santos and Veronesi (2022), the fact that the long-run consumption risk correlates with the prediction of the intermediary-based model does not indicate which shock causes the other. To ascertain whether consumption or intermediary capital is the fundamental source of shocks that are priced in the cross-section of corporate bonds, we need to go beyond the reduced-form analysis presented in this article.

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Endnotes

¹ See Goldberg and Nozawa (2021) for the evidence of intermediary affecting bond liquidity and risk premiums.

² The CEX is a nationwide household survey conducted by the U.S. Bureau of Labor Statistics designed to provide detailed data on spending, income, demographics, and asset holding information. The data are publicly available at <https://www.bls.gov/cecx/>.

³ We also use the most recent samples of 2020 and find that our main results are robust to the extended sample periods in which the consumption level significantly drops because of the pandemic. This result is reported in Table IA1 in the online appendix.

⁴ We impose the minimum number of observations at 24 months.

⁵ To compute idiosyncratic volatility, we run regressions of bond returns on the five factors of Fama and French (2015) over the 36-month rolling window (with the minimum number of observations of 24 months) and calculate the standard deviation of the regression residuals.

⁶ We do not use short-term reversal, as Bai et al. (2019) show that it is not a source of systemic risk and likely to be driven by the market microstructure noise in the data.

⁷ We also estimate the betas over the 36-month rolling windows, but this method does not lead to a significant difference in average returns between the first and last quintiles; thus, we use the portfolios based on 12-month rolling window betas as our main results.

⁸ The exact starting month varies across portfolios. The credit rating and maturity portfolios start in February 1973, the credit spread portfolios start in April 1973, intermediary factor beta portfolios start in February 1974, downside risk and idiosyncratic volatility portfolios start in February 1975, and long-term reversal portfolios start in February 1977.

⁹ See, for example, Ingersoll (1987).

¹⁰ In Online Appendix III, we calibrate the long-run risk model of Bhamra et al. (2010b) and show that long-run consumption risk is

indeed an important driver of expected returns on corporate bonds in addition to credit spreads.

¹¹ The GMM moment conditions are

$$E \begin{bmatrix} r_{i,t+1} - r_{f,t} + \frac{\sigma^2(r_{i,t+1}) - \sigma^2(r_{f,t})}{2} - \zeta - (\gamma - 1) \\ \varepsilon_{c,t \rightarrow t+S}(r_{i,t+1} - r_{f,t}) \\ \varepsilon_{c,t \rightarrow t+S} \end{bmatrix} = 0,$$

where $\varepsilon_{c,t \rightarrow t+S} = \sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s}) - \mu_{\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s})}$, $\theta = (\zeta \ \gamma \ \mu_{\sum_{s=0}^{S-1} \delta^s (c_{t+1+s} - c_{t+s})})'$ are the parameters to be estimated.

Because there are 41 moment conditions in total and three free parameters, the GMM procedure finds the best parameters to minimize the sum of squared moments

$$\min_{\theta} g_T(\theta)' W g_T(\theta),$$

where $g_T(\theta)$ is the sample counterpart of the moments in the first equation and W is the weighting matrix defined by

$$W = \begin{pmatrix} I_{40} & 0 \\ 0 & H \end{pmatrix},$$

and we set H to be the number of test assets, which is sufficiently large to ensure that the mean of the long-run consumption growth rate is well measured.

¹² The stationary bootstrap procedure introduced by Politis and Romano (1994) is used with the random block lengths drawn from a geometric distribution to ensure the stationarity of the resulting time series. Specifically, we resample blocks of asset returns, risk-free rate, and the long-run risk measures randomly with replacement until the bootstrap sample size is equal to the number of real data observations. Then, we obtain estimates and $\bar{R}^2$ by rerunning the regression using the bootstrap samples. We repeat this procedure 5,000 times and construct the bootstrap distribution of estimates and $\bar{R}^2$. To choose the expected block length, we follow Politis and White (2004) and set the optimal expected block length.

¹³ Although the estimated level of risk aversion $\gamma = 15$ is more reasonable than those provided in the literature (as summarized in Table IA2 in the online appendix), it is still higher than the conventional level in an interval of 0.5–2. To achieve even lower values, we need additional features in the model. For example, Collin-Dufresne et al. (2016) show that an economy where an agent faces unknown model parameters and has to learn them using observed states generates high risk premiums for risky assets. Hansen and Miao (2022) extend the model of Hansen and Sargent (2011), where an agent is averse to model ambiguity, and show that this additional uncertainty reduces investor demand for risky assets. These additional model features may help reduce the risk aversion in the paper even further. However, we do not take this path to emphasize the simplicity and transparency of the empirical analysis.

¹⁴ Malloy et al. (2009) report the cross-sectional $\bar{R}^2$ of 0.65 for the 25 size- and book-to-market-sorted portfolios using 24-quarter shareholder consumption growth and 0.52 using 24-quarter top shareholder consumption growth (see table II in Malloy et al. 2009).

¹⁵ Furthermore, Kroencke (2017) points out that NIPA statisticians filter observable consumption, and this leads to seemingly too smoothed NIPA consumption measures.

¹⁶ Kaplan and Violante (2014) show that some wealthy households behave as if they were constrained when their assets are in illiquid assets, such as real estate and retirement accounts. We argue that wealthy households in our setup are unlikely to be constrained as we measure assets using stocks, mutual funds, and bonds rather than real estate. Furthermore, we focus on the top 30% of wealthy households to avoid constrained households.

¹⁷ We use the fitted treasury yield curve of Gurkaynak et al. (2006) available on the Federal Reserve's website and discount corporate bond cash flows using treasury zero-coupon rates to obtain a price of matching treasury bonds. Specifically, we use the parameter of the Svensson (1994) model available in the data set and calculate the zero-coupon rate for the exact payment date for each coupon and principal for each corporate bond.

¹⁸ As in Malloy et al. (2009), we exploit the entire time series of returns starting from July 1926 to precisely estimate average returns. In order to obtain the bootstrap sample size that is equal to the number of real data observations for both the left-hand side (LHS) sample and the right-hand side (RHS) sample for this case, bootstrapping is performed in the following way. First, we place two sets of the data separately to be resampled: (1) the time period where only the LHS is available and (2) the time period where both the LHS and RHS are available. From each bootstrap sample, we attach the resampled first data set to the corresponding resampled second data set for each asset to obtain a bootstrap sample for the LHS. A bootstrap sample for the RHS is obtained directly from the resampled second data set.

¹⁹ For further robustness, in Online Appendix V, we show that the results in Panel A are robust when we estimate the regression in (14) in reverse, switching the LHS variable and RHS variable.

²⁰ In Online Appendix VI, we repeat the exercise using betas (rather than covariance) for the long-run consumption growth rate and show that the fit of the model and estimated price of risk are consistent with the main results.

²¹ The 128 variables are from McCracken and Ng (2016).

²² To support this claim, we show in panel B of Table 7 that the selected factors generally do not exhibit the same predictive power for asset returns as for consumption growth. This implies that the quantity of risk using this VAR approach will not be mechanically led by a high predictive power of the selected factors for bond returns.

²³ Table IA6 in the online appendix presents the summary statistics of shocks to the expectation for long-run consumption growth estimated using the VAR.

²⁴ The constant term in (18) is $E[(1 - \gamma)\varepsilon_{c,t \rightarrow t+S}] + 1/2E[(1 - \gamma)^2 \varepsilon_{c,t \rightarrow t+S}^2] + 1/6E[(1 - \gamma)^3 \varepsilon_{c,t \rightarrow t+S}^3] + 1/6E[r_{f,t+1}^3]$. Because the last term is small, we assume that $E[r_{f,t+1}^3] = E[r_{f,t}^3]$ in deriving (20).

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